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**Are the Day of the week effects prevalent in Indian Stock
Exchanges? A COVID-19 Contextual Analysis**

Student Name: Nehru Hari

Student Number: 2383228

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Abstract:

This dissertation investigates the presence and persistence of day-of-the-week effects in the Indian equity market, focusing on the Nifty50 and Sensex indices from the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE). Using daily data from January 2018 to January 2023, the study examines whether index returns systematically differs across weekdays, with particular attention to changes during the COVID-19 pandemic. These anomalies, if present, challenge the weak-form Efficient Market Hypothesis (EMH), which asserts that asset prices fully reflect past return information. A multi-stage econometric approach is adopted to investigate this Anomaly, Ordinary Least Squares (OLS) and Autoregressive (AR) models are first employed to test for weekday return differentials, A structural break analysis on a date declared as an international emergency due to the outbreak of coronavirus disease assesses shifts in return behaviour pre- and post-pandemic, after this Generalized Auto Regressive Conditional Heteroskedasticity (GARCH), Exponential GARCH, and Threshold GARCH models capture time-varying volatility and test for weekday effects in volatility, Later, an Amihud-based illiquidity measure is used to explore whether liquidity influences return patterns, and Finally ARIMA-based regressions are used to evaluate EMH by testing return predictability. Results reveal statistically significant Tuesday effects in both indices, with heightened returns especially during the pandemic. Structural break regressions confirm a shift in return dynamics post-COVID. Volatility modelling finds evidence of conditional heteroskedasticity and leverage effects, though weekday dummies in volatility equations are often insignificant. Volume and illiquidity factors, particularly in the BSE, partially explain weekday anomalies. EMH testing detects mild autocorrelation, suggesting limited predictability in returns. In conclusion, while weekday effects are observed, they are not consistently strong or persistent. The findings indicate partial inefficiencies, particularly during the pandemic, this may overall support a trend toward increased efficiency in Indian equity markets.

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Author's Declaration:

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material, which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

SIGNED:

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1. Introduction:

In recent decades, the exploration of market anomalies has emerged as a cornerstone of empirical finance, reshaping how researchers and practitioners understand market behaviour. Market Anomalies refer to deviations in asset return patterns which challenge the Efficient Market Hypothesis (The theory that all available information is fully reflected in asset prices, making it impossible to consistently achieve above-average returns). These anomalies indicate instances where securities appear to be systematically and predictably mispriced, implying that markets are not consistently efficient. Such anomalies may occur due to investor behaviour, like an emotional or irrational decision-making, delays in how information is shared, limitations within institutions, or risks that traditional pricing models fail to consider. Common examples include calendar-based patterns like the January effect (Stocks tend to rise more in January than in other months of the year) or day-of-the-week effect (average daily returns tend to differ across weekdays) which impacts the size effect (which occurs when small-cap companies beat large-cap stocks), value effect (when undervalued stocks create greater returns), momentum effect (where previously winning stocks continue to perform well in the market), and so on.

The day-of-the-week effect is a statistically significant and consistent variation in stock returns that is contingent upon the day of the week. Studies have indicated that average returns are typically abnormally low on Mondays and exceptionally high on Fridays, revealing that the market doesn't fluctuate randomly, but it follows a predictable sequence of stages or phases. This anomaly has been detected across several markets, encompassing both established and emerging economies. Furthermore, some studies have indicated fluctuations in volatility throughout weekdays, implying that risk levels may vary during the week. These results can be significantly important to investors because it demonstrates that the returns they earn and the risks they face can depend on which day of the week they invest.

Cross (1973) was the first to formally record the day-of-the-week effects; Later, French (1980) popularized the concept after discovering that U.S stock returns were substantially lower on Mondays and higher on Fridays. This paradoxical outcome created significant interest among practitioners and academics, resulting in several extension studies spanning from different timeframes to international financial markets. Gibbons and Hess (1981) validated the Monday return anomaly and presented preliminary econometric evidence attesting to its importance. Keim and Stambaugh (1984) further investigated this phenomenon utilizing cross-sectional

data, attributing it in part of microstructure- related factors like bid-ask spreads. Lakonishok and Levi (1982) posited that settlement procedures and trading delays can contribute to the anomaly. In contrast, Rogalski (1984) contended that negative Monday returns are predominantly caused by returns over the weekend, which led to discussions of the weekend effect as a subset of the broader day-of-the-week pattern. Furthermore, Damodaran (1989) investigated these effects concerning risk premia and behavioral rationales. Numerous explanations were proposed by these researchers, including delays in corporate news releases, measurement error, bid-ask spread effects, and trader behavior during non-trading sessions. While these explanations make sense in theory, most empirical studies have found weak or no evidence of them. Such discrepancy in finding the results highlights the difficulties inherent in investigating the day-of-the-week impact. A major challenge is data mining, wherein repeated testing across many timeframes, marketplaces, and model parameters can yield results that appear substantial but lack economic validity. Schwert (2003) noted that numerous abnormalities may merely be artefacts of extensive empirical investigation rather than genuine inefficiencies. Furthermore, the evolution of financial markets over time-encompassing enhancements in trading technology, increased market efficiency, and evolving investor behavior- can diminish or modify previously identified patterns, complicating the derivation of consistent conclusions across various periods. Structural disparities among markets, involving trading hours, settlement procedures, and liquidity levels, exacerbate the complexity of cross-market comparisons. Moreover, the convergence of calendar anomalies, including holidays, frequently complicates the isolation of the intrinsic day of the week effect. The lack of a universally recognized theoretical framework for this anomaly hinders the creation of reliable predictive models, leaving much of the existing literature open to contention and reanalysis.

Connolly (1989, 1991) introduced an additional degree of complexity by challenging the validity of initial findings, by demonstrating that the significance of the day-of-the-week effect declines when statistical corrections for heteroskedasticity, autocorrelation, and normality are implemented. This raised concerns about spurious results and data mining (Sullivan et al., 2001; Alt et al., 2011), prompting researchers to adopt more robust econometric techniques. Tools like Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models help account for time-varying volatility, while robust standard errors, Newey-West corrections, and bootstrap methods enhance result reliability. Additionally, generalized F-tests and closure tests are used to address the risks of multiple hypothesis testing.

Even with these developments, the subject of market anomalies continues to remain controversial. Critics argue that this could be due to factors like data snooping (Statistical bias that occurs when researchers repeatedly analyse the same dataset to find patterns, often leading to the discovery of spurious correlations or relationships that are not truly significant), restricted sample periods, or structural changes in markets over time—a phenomenon known as the "anomaly fade effect." As markets evolve with new technologies, smarter investors, and better access to information, such inefficiencies may be quickly arbitrated away. Fama (1998) and Schwert (2003) suggest that many anomalies may be due to chance and often lose relevance post-publication. Sullivan et al. (2001) further note that adjusting for multiple testing reduces the statistical significance of many calendar effects.

Although the anomaly may have diminished in developed markets such as the United States of America. There is notable evidence of these anomalies in emerging markets. According to Jaffe and Westerfield (1985), the pattern of day-specific returns is different in different countries. Suggesting that these effects are not happening simultaneously everywhere and may be affected by investor behavior and institutional frameworks. Thus, prompting scholars to redirect their attention to non-U.S. markets, where information asymmetries and market frictions are more significant.

Emerging markets provide an intriguing framework for analyzing the persistence of the day-of-the-week effect. In contrast to their developed counterparts, these markets frequently exhibit greater information asymmetries, diminished regulatory control, reduced liquidity, and more pronounced investor behaviors. Such characteristics can result in increased pricing inefficiencies, creating a thriving environment for calendar anomalies. Recent studies underscore the necessity of exploring these anomalies in emerging markets, characterized by structural inefficiencies, diminished liquidity, and heightened behavioural biases. Researchers like Bower & Dimson (1988) and Lakonishok & Smidt (1988) have suggested looking at these effects in different countries to find out if calendar anomalies happen everywhere or are just due to specific local conditions.

This study narrows down the focus to one geographic location of the emerging economy: India. Historically, India's capital markets have evolved significantly since the 1991 liberalisation reforms—from heavily regulated, fragmented systems to highly digitised, globally integrated platforms. Today, India is home to two of Asia's largest stock exchanges, the Bombay Stock Exchange (BSE) and National Stock Exchange (NSE), supported by a robust regulatory

framework under the Securities and Exchange Board of India (SEBI). Geopolitically, India's influence continues to grow across the Indo-Pacific, strengthening its position in global trade, investment, and economic policy discourse. The country has witnessed a surge in domestic and foreign investor participation, bolstered by structural reforms, digital adoption, and increasing representation in global indices like MSCI, with imminent inclusion in sovereign bond indices. As a result, India has become a focal point for global investors seeking exposure to high-growth markets, which makes India a feasible topic of interest to this study.

Furthermore, this study focuses on investigating the day-of-the-week anomaly in the Indian stock market within the unique context of the COVID-19 era. By focusing on this unique period of economic uncertainty, this research aims to identify if there are any specific shifts in return patterns—such as the reoccurring of negative Mondays returns, stronger Tuesday or Friday effects, and if there was any increased volatility caused by investor reactions to weekend news. These findings could reveal a structural break in how returns behave and show how investor sentiment, trading activity, and risk perception changed during the pandemic. Understanding these shifts is crucial for developing smarter trading strategies, improving risk management, and guiding policy decisions in one of the world's most important emerging financial markets.

This dissertation analyses the continuity and progression of the day-of-the-week (DOW) effect in Indian stock exchanges, utilising daily return data from two principal indices: Nifty 50 (NSE) and Sensex (BSE). It seeks to offer a unique perspective by analysing the anomaly within the framework of the COVID-19 pandemic. Analysis revisits daily return patterns based on the paradigm established by Khan et al. (2021), but utilizing recent data from a period characterized by significant economic uncertainty and market disruption. To thoroughly examine the DOW effect, this research builds upon the econometric advancements discussed above, such as time-series models that account for autocorrelation and heteroskedasticity. This study employs advanced return and volatility modelling techniques that account for time-varying risk and market behaviour asymmetries to identify average return differences across weekdays and analyse volatility dynamics. The methodology incorporates structural break analysis to assess the impact of COVID-19 on return patterns, alongside volume-based regressions that utilise illiquidity proxies to examine microstructural influences. Furthermore, evaluations of market efficiency via return predictability models will determine the degree to which these anomalies challenge the weak-form Efficient Market Hypothesis.

This dissertation is structured as follows: Section 2 reviews existing literature on the day-of-the-week anomaly in both global and Indian markets; Section 3 describes the type of data and methodological framework utilised to analyse weekday return differentials and volatility patterns; Section 4 presents the results of empirical findings; Section 5 describes the discussions of the findings obtained; and Section 6 is the conclusion.

2. Literature Review

2.1 Global Evidence of the day-of-the-week effect:

Numerous studies have explored the persistence of day-of-the-week (DOW) effects in global markets. A key early contribution by Choudhry (2000) examined seven emerging Asian markets—India, Indonesia, Malaysia, the Philippines, South Korea, Taiwan, and Thailand—using GARCH models from 1990 to 1995. The study found significant weekday anomalies in most countries, particularly negative Monday returns in Malaysia and Thailand. India, however, showed weak or no DOW effects. Volatility was also found to vary across the week, suggesting that return anomalies may align with shifts in market risk. Choudhry also suggested that spillovers from developed markets like Japan could help explain these patterns. However, the study is limited by its older dataset and lack of structural break or volume-based analysis, which could have better captured evolving market conditions.

Khan et al. (2021) provided a more updated and detailed analysis using data from 2013–2019 across eleven emerging Asian markets. Using OLS, GARCH (1,1), and non-parametric Kruskal-Wallis tests, they confirmed weekday effects in countries such as China, Indonesia, and Pakistan, while finding no significant effects in India and Malaysia. The study uniquely incorporated traded volume to examine whether liquidity influences these anomalies. Although Mondays were generally the least traded day, the link between low volume and abnormal returns was inconsistent. This research importantly added a microstructure perspective but did not consider structural events like economic crises.

A broader global view comes from Chang, Pinegar, and Ravichandran (1993), who analyzed 22 markets using FT-Actuaries World Index data. They applied robustness checks such as Bayesian t-tests and heteroskedasticity-robust errors. Their findings showed that while DOW effects disappeared in the U.S. after corrections, they remained strong in several European markets. Interestingly, anomalies were not consistent across months, suggesting they may not be stable over time. This study emphasized the need for strong econometric techniques and market-specific analysis when studying DOW effects.

Taken together, these studies reveal three key insights relevant to this dissertation. First, DOW effects are not universal—they differ substantially across countries and periods, shaped by institutional structures, investor behaviour, and possibly regional spillovers. Second, empirical validity is contingent on methodological rigor; studies employing GARCH models or adjusting

for heteroskedasticity and autocorrelation provide more reliable inference. Third, the role of liquidity and structural events remains underexplored—creating a research gap addressed in this dissertation by integrating volume analysis and structural break tests during the COVID-19 period.

2.2 Day of the Week effect from the Indian context:

Empirical studies on the day-of-the-week (DOW) effect in Indian stock markets have presented a rich but mixed body of evidence. One of the earliest comprehensive studies by Bhattacharya et al. (2003) examined data from 1991–2000 using BSE-100 index returns. They applied GARCH models with week-level dummy variables to account for reporting and non-reporting weeks. The findings revealed significant positive returns on non-reporting Thursdays and Fridays, suggesting that institutional settlement cycles influenced daily return patterns. Their subperiod analysis further indicated changes in DOW effects post-1995, reflecting structural shifts such as NSE's emergence and dematerialization. While their tailored approach is good, the lack of asymmetry modelling and the dated nature of their data limit its relevance for more recent market dynamics.

Srinivasan and Kalaivani (2013) extended this research by analyzing a broader time frame (1997–2012) for NSE and BSE using GARCH(1,1), EGARCH(1,1), and TGARCH(1,1) models. Their results confirmed the presence of positive Monday and Wednesday effects in both indices, while volatility analysis revealed that Tuesday had a negative and significant impact. Their use of asymmetric GARCH models helped uncover leverage effects in Indian markets, underscoring the non-linear and information-sensitive nature of return volatility. Despite the robust econometric framework, the study ends before the COVID-19 period and does not address how crises may revive or reshape anomaly patterns.

A later contribution by Nagpal and Jain (2018) investigated the weak-form efficiency of Indian markets using Nifty 50 data from 2000–2017. They applied OLS and EGARCH models and identified a significant Wednesday effect. Their use of breakpoint unit root testing adds a structural lens, showing how crises like the 2008 financial meltdown may influence return distributions. While their methodology is solid and builds upon earlier findings, it lacks focus on trading volume or investor types, which are important when explaining anomalies from a microstructural perspective.

Sahoo (2021) contributed a significant update by focusing on six Nifty indices before and during the COVID-19 pandemic. Using dummy variable regression and GARCH modelling, the study found no significant DOW effects pre-COVID, but clear anomalies emerged during the pandemic—specifically, negative Monday returns and strong Tuesday and Friday effects. EGARCH results showed significant asymmetric volatility and volatility clustering, which align with broader global trends during crisis periods. This study is crucial in highlighting how calendar effects may reappear during market stress, contradicting assumptions of stability in market efficiency.

Aggarwal and Jha (2023) used GARCH, EGARCH, and TGARCH models on NSE-Nifty data from 1990 to 2022. The study identified significant and positive returns across all weekdays, with Monday and Wednesday being most profitable. Asymmetric volatility was evident, particularly with Tuesday showing a negative effect on conditional variance. The long-time span, combined with the use of multiple GARCH variants, makes this study especially relevant. However, it does not isolate crisis periods like COVID-19 or incorporate volume-based analysis.

The most recent and methodologically comprehensive study by Yadav and Kadian (2025) also focused on the pandemic period (March 2020–September 2021), examining DOW effects using Sensex, Nifty 50, and SX-40 data. Their dummy regression model found negative Monday returns (significant only for Sensex) and significant positive effects on Tuesday, Wednesday, and Friday. They employed robustness checks such as the Durbin-Watson and Breusch-Pagan tests. While the methodology is rigorous, it does not apply GARCH-family models, thus missing key insights on volatility behaviour.

2.3 Motivation:

This study is driven by the persistent academic and practical relevance of day-of-the-week (DOW) anomalies and the need to reassess their presence under contemporary and extreme economic conditions, particularly in a rapidly evolving market like India. Despite a rich body of literature investigating DOW effects globally, the Indian context remains underexplored in recent years—especially under the disruptive impact of the COVID-19 pandemic. While several earlier studies use robust GARCH-family models, few consider structural disruptions or integrate volume-based explanations for weekday anomalies. Moreover, empirical findings across Indian indices have been inconsistent, varying by time period, index, and methodology.

These limitations highlight the need for updated analysis using current data and multi-model testing frameworks.

The Indian equity market has undergone significant transformation—driven by digitalisation, increased retail participation, and macroeconomic reform. These shifts may simultaneously improve market depth while introducing behavioural inefficiencies. Understanding how these changes manifest across trading days is crucial. The selection of Sensex and Nifty50 ensures market-wide relevance due to their depth, liquidity, and role as barometers of investor sentiment.

COVID-19 presents a compelling natural experiment for examining the resilience and evolution of such anomalies. The pandemic introduced unprecedented volatility, uncertainty, and behavioural shifts among investors. Analyzing whether anomalies such as the Tuesday effect re-emerged, intensified, or diminished post-pandemic helps to determine how systemic shocks reshape market dynamics.

Furthermore, recent Indian studies often neglect illiquidity and volatility asymmetries. By incorporating Amihud illiquidity ratios and structural break regressions, this study could bridge a gap in understanding how market microstructure variables interact with weekday patterns.

Overall, this research aims to address three critical gaps: (1) the persistence of DOW anomalies during systemic shocks, (2) the role of volume and illiquidity in explaining these patterns, and (3) the changing volatility dynamics captured through asymmetric GARCH models.

3. Data and Methodology:

3.1 Data:

This study utilises a five-year time series dataset of daily closing prices, spanning from 1-January-2018, to 1-January-2023, to analyse the influence of day-of-the-week effects on the Indian market. The dataset comprises the Bombay Stock Exchange's Sensex index and the National Stock Exchange's NIFTY50 index, which are two of the prominent stock exchanges and serve as significant indicators of the Indian economy, both these datasets were sourced from the Refinitiv workspace of LSEG. The Sensex index of the Bombay Stock Exchange (BSE) consists of thirty reputable and financially robust companies with a significant market presence in India. Additionally, the National Stock Exchange's NIFTY50 index includes fifty of the most prominent companies traded in the Indian equities market. Moreover, both stock exchanges are value-weighted indices. Both Sensex and NIFTY50 are consistently monitored by retail investors, forex traders, institutional investors, and foreign institutional investors. These indices are widely recognised and actively observed by global investors for trading purposes. Although Sensex has a longer historical presence than the National Stock Exchange, NIFTY50 has emerged as a favoured trading choice since the twentieth century. The examination of day-of-the-week effects during the COVID-19 period is suitable, as these datasets are relevant for such analysis. Previous studies have utilised these datasets to investigate these effects. Conducting research on these indices provides a valid parameter for assessing the existence of day-of-the-week effects, the efficient markets hypothesis, the influence of volume on returns, and parts of volatility modelling.

Several variables are of importance when investigating for the day of the week effect. Table 1 depicts all the variables and their meaning in context of the study. The return variable is estimated by the ratio of the logged value of present prices to its previous close prices as shown in equation (1):

$$Return_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \quad (1)$$

Where, P_t denotes the closing price on day t , and P_{t-1} on the previous trading day.

Variables	Units	Descriptions
Price	Indian National Rupee	Closing price of the index on a particular day (Quantitative-Continuous variable)
Return	-	Logged ratio of price of closing price of present day to its closing price of previous day (Quantitative-Continuous variable)
Volume	Number of shares (index) traded	Total number of shares traded on a given trading day
Mon	-	Dummy variable indicating the presence of closing price on a Monday (Categorical Variables)
Tue	-	Dummy variable indicating the presence of closing price on a Tuesday (Categorical Variables)
Wed	-	Dummy variable indicating the presence of closing price on a Wednesday (Categorical Variables)
Thu	-	Dummy variable indicating the presence of closing price on a Thursday (Categorical Variables)
Fri	-	Dummy variable indicating the presence of closing price on a Friday (Categorical Variables)
Timeid	-	Describes the number of entities for the time series relationship (Quantitative-Discrete variable)
Date	-	Describes the date represented in the Stata software (Categorical-Continuous variable)

Table 1: Describes the variables and their description used in the study.

The total sample size for Sensex data is 1237 data points indicating closing prices for 1237 trading days over the 5-year period, whilst the NIFTY50 had 1238 trading days over the same period. Table 2 and Table 3 present the descriptive statistics of Sensex and NIFTY50 indices.

Variables	Observations	Mean	Std. dev.	Min	Max
price	1237	44695.1	9817.438	25981.24	63284.19
return	1237	0.000475	0.012485	-0.14102	0.085947
volume	1237	16500000	12100000	1835714	1.88E+08
mon	1237	0.200485	0.400525	0	1
tue	1237	0.199677	0.399919	0	1
wed	1237	0.199677	0.399919	0	1
thu	1237	0.201294	0.401129	0	1
fri	1237	0.197251	0.398085	0	1
timeid	1237	620	357.2354	2	1238
date	1237	22100.45	527.5484	21186	23009

Table 2: Descriptive statistics of BSE's Sensex dataset.

Variables	Observations	Mean	Std. dev.	Min	Max
price	1238	13327.23	2909.114	7610.25	18812.5
return	1238	0.000445	0.012331	-0.13904	0.084003
volume	1238	4.26E+08	2.14E+08	2826000	1.81E+09
mon	1238	0.200323	0.400404	0	1
tue	1238	0.199515	0.399798	0	1
wed	1238	0.199515	0.399798	0	1
thu	1238	0.201104	0.401008	0	1
fri	1238	0.197092	0.397963	0	1
timeid	1238	620.5	357.5241	1	1239
date	1238	22100.24	527.3835	21186	23009

Table 3: Descriptive statistics of NSE's NIFTY50 dataset.

The NSE exhibits a superior average volume of ₹426 million, in contrast to the BSE's ₹16.5 million, signifying enhanced trading activity. The mean daily returns for both indices are approximately same, however the BSE exhibits marginally greater return volatility. The distribution of weekday dummy variables shows that returns are evenly spread across all five trading days, with each day (Monday to Friday) accounting for roughly 20% of the week's overall return activity. Both datasets exhibit structural similarities yet demonstrate divergent market liquidity and trade scale.

3.2 Methodology:

This section explains all the sophisticated time series models that are used to investigate for the day of the week effects.

3.2.1 OLS Approach:

The first phase of this study utilizes the Ordinary Least Squares (OLS) regression technique to examine the existence of day-of-the-week's effect in the returns of NIFTY50 and SENSEX indices. The primary aim is to ascertain if average stock returns fluctuate considerably among several weekdays. OLS is used to establish a baseline understanding of weekday return behaviour, allowing for a clear and interpretable comparison of average returns across weekdays which is critical for identifying the presence of calendar anomalies before applying more advanced models.

3.2.1.1 Model Specification

The multiple linear regression model is constructed (as shown in Equation (2) below) which includes dummy variables for weekdays—Monday through Thursday—while Friday is excluded and serves as the base category, additionally, such an approach ensures that we avoid the "dummy variable trap," a form of perfect multicollinearity that occurs when redundant variables are included in a regression. By using only four dummies instead of five, the model remains identifiable and interpretable. It is also important to note that Equation (2) serves as the base case model for this dissertation.

$$\text{Return}_t = \alpha + \beta_1 \text{Mon}_t + \beta_2 \text{Tue}_t + \beta_3 \text{Wed}_t + \beta_4 \text{Thu}_t + \varepsilon_t \quad (2)$$

In Equation (2), Return_t represents the daily return of the stock index on day t . The dummy variables Mon_t , Tue_t , Wed_t , and Thu_t take the value "1" if the return corresponds to that specific weekday and "0" otherwise, with Friday omitted as the base category. The coefficients β_1 to β_4 capture the average return difference for each weekday relative to Friday. α is the intercept, representing the mean return on Fridays, while ε_t is the error term accounting for unexplained variations in returns.

Equation (2) enables the comparison of mean returns on each weekday against Friday. If any of the coefficients β_i differ significantly from zero, it implies that return behaviour varies by weekday, suggesting a violation of the weak-form Efficient Market Hypothesis (EMH).

To ensure robustness of inference, diagnostics tests are performed:

- To evaluate weekday return differences, t-tests are conducted on individual β coefficients in the OLS model. Each test assesses whether returns for a specific weekday significantly differ from Friday, the base category. The t-statistic is calculated by dividing the estimated coefficient by its standard error. A p-value below 0.05 typically leads to rejection of the null hypothesis, indicating a statistically significant deviation in returns for that day.
- In addition, an F-test examines whether all weekday dummy coefficients (Monday through Thursday) are jointly zero. This test evaluates the presence of a day-of-the-week effect. A significant F-statistic implies that at least one weekday return differs from Friday, confirming systematic variation in return behaviour across the week.
- To validate model assumptions, residual diagnostics are performed. The Breusch-Godfrey (BG) test checks for autocorrelation in regression residuals. Since serial correlation can bias standard errors and mislead inference, this test regresses residuals

on lagged residuals and predictors. If autocorrelation is detected, adjustments such as Newey-West standard errors are applied. These heteroskedasticity- and autocorrelation-consistent (HAC) standard errors use weighted residual autocovariances to produce valid inference under time-varying volatility conditions.

- Finally, the Augmented Dickey-Fuller (ADF) test is used to confirm stationarity in the return series. Stationary data ensures reliable regression results. The test evaluates the presence of a unit root, with a significant negative statistic rejecting the null and confirming the stationarity required for accurate time-series analysis.

3.2.1.2 Lag Selection and Model Refinement:

Given the time series nature of the data, lagged returns were added (as shown below in Equation (3)) to control for potential autocorrelation in returns:

$$\text{Return}_t = \alpha + \beta_1 \text{Mon}_t + \beta_2 \text{Tue}_t + \beta_3 \text{Wed}_t + \beta_4 \text{Thu}_t + \phi_1 \text{Return}_{t-1} + \varepsilon_t \quad (3)$$

In Equation (3), ϕ_1 is the coefficient associated with the lagged return term (Return_{t-1}), hence, the term $\phi_1 \text{Return}_{t-1}$ introduces a lag of the previous day's return to control for autocorrelation; the other variable descriptions are the same as those explained in the base case regression from Equation (2).

Including lagged returns helps distinguish true weekday effects from serial correlation in returns, ensuring that any detected anomalies are not falsely attributed to autocorrelation patterns inherent in financial time series.

Lag order is set depending on information factors like the Akaike Information Criterion (AIC). A detailed explanation of this selection process is provided in the subsequent volatility modelling section.

The OLS model will be executed independently for the entire sample and for specific sub-periods — pre-COVID (2018–2019), during COVID (2020), and post-COVID (2021–2022) — to identify potential alterations in behavioural anomalies attributable to the epidemic. It is important to note that the pre-COVID and post-COVID regressions have two years of data samples as opposed to one year of sample of data for the during COVID regressions. These distinct regressions would help in observing if there was any statistically significant day of the week dummies, this key information could help in finding if any patterns for the different day of the week effect emerged, persisted or diminished during the period specified.

3.2.2 Structural Break Testing:

Financial time series data, such as stock market returns, frequently shows changes in the underlying data-generation process because of important economic or geopolitical events. These changes are known as structural breakdowns, they arise due to rapid variations in the relationship between variables across time that, if not accounted for these factors may invalidate typical time series models. Structural break analysis enables this study to test whether the unprecedented shock of COVID-19 permanently altered weekday return behaviour, which is essential for assessing the temporal stability of DOW effects in a crisis-driven environment. Ignoring structural breaks can result in inaccurate parameter estimations and faulty inference.

This study selects 31 January 2020 as the structural break point, corresponding to the WHO's declaration of COVID-19 as a global public health emergency. This date was marked as the first formal global response to the pandemic and triggered early volatility in financial markets, making it a logical choice to assess shifts in return behaviour. While later events, such as India's nationwide lockdown on 25 March 2020 and the vaccine rollout beginning on 16 January 2021, may also reflect structural changes, incorporating multiple breakpoints is beyond the scope of this dissertation due to model complexity and limited sub-sample sizes. Limiting the analysis to a single, globally significant event could ensure clarity, tractability, and consistency across econometric models.

The goal is to see if the day-of-the-week effects (patterns in daily returns linked with specific weekdays) alter in magnitude or direction before and after this structural breakpoint. The analysis tries to identify anomalies in market efficiency by separating the data into two periods: pre-COVID (01/01/2018-30/01/2020) and post-COVID (31/01/2020-30/12/2022).

3.2.2.1 Model Specification with Dummy variables and interaction term approach:

To proceed with structural break testing, a pooled regression with interaction terms is used to formally test for the presence of a structural break. The interaction term approach specifically identifies how weekday effects shifted post-COVID, allowing for a differential analysis of return patterns that reflects regime changes—something simple pre/post comparison models

cannot detect. A dummy variable “COVID” is created to equal the value of “1” for dates from 31st January 2020 and “0” otherwise.

The model is represented as follows:

$$R_t = \alpha + \beta_1 D_{\text{Mon}} + \beta_2 D_{\text{Tue}} + \beta_3 D_{\text{Wed}} + \beta_4 D_{\text{Thu}} + \gamma D_{\text{Covid}} + \delta_1 (D_{\text{Mon}} \times D_{\text{Covid}}) + \dots + \delta_4 (D_{\text{Thu}} \times D_{\text{Covid}}) + \varepsilon_t \quad (4)$$

Here:

- The δ_i terms capture how the day-of-the-week effects change in the post-COVID period relative to the pre-COVID period.
- R_t represents the daily return on day t .
- β_1 to β_4 capture the average return difference for each weekday relative to Friday.
- D_{Mon} to D_{Thu} , are dummy variables that indicate whether the day is Monday to Thursday.
- γ is the coefficient for D_{Covid} dummy variable.
- D_{Covid} is a dummy variable equal to “1” indicating the post-COVID structural break period and “0” otherwise.
- α is the intercept (average return on Fridays pre-COVID), and ε_t is the error term.

The importance of the interaction terms is again evaluated by individual t-tests, and a joint F-test is conducted to ascertain whether the set of interactions is jointly significant. A higher value of F-statistic would indicate that the day-of-the-week effects are inconsistent between the two periods, thereby proving a structural break.

3.2.2.2 Chow Test for Structural Break:

The Chow Test is an official technique used to evaluate structural stability. In this study, it helps assess whether the relationship between weekday returns and market performance shifted due to COVID-19. By comparing regressions run on the full sample with those from the pre- and post-COVID periods, this test would help reveal whether a structural break exists or not. The Chow Test is essentially a specialised form of the F-test, its F-statistic is calculated using the formula shown in Equation (5):

$$F = \frac{(SSR_p - (SSR_1 + SSR_2))/k}{(SSR_1 + SSR_2)/(n_1 + n_2 - 2k)} \quad (5)$$

Where:

- SSR_p is the sum of squared residuals from the full sample.
- SSR_1, SSR_2 are sum of squared residuals from the pre- and post-COVID periods.
- k is the number of parameters.
- n_1 and n_2 are the number of observations in each period.

If the calculated F-statistic surpasses the critical value, the null hypothesis of structural stability is dismissed, signifying a structural break at the designated point. Alternatively, we can also look at the p-value of this test to base a conclusion on the existence of structural break, generally a p-value below 0.05 would mean the null hypothesis is rejected and suggesting a significant difference between regression models, and thus a structural break.

3.2.3 Volatility Modelling:

This part specifies the use of time-series volatility modelling approaches performed to assess the return behaviour in the two exchanges in study, transcending conventional typical return patterns. These strategies are particularly suitable when returns exhibit time-varying volatility, which is a common occurrence in financial time series data, wherein periods of high volatility are typically succeeded from large volatility periods, and conversely.

Modelling return volatility is crucial in this investigation as it enhances comprehension of risk dynamics inside the Indian stock markets, especially regarding the anomalies examined. Conventional mean-based regressions, such as OLS, can show average return patterns across weekdays; but they neglect the unpredictability and clustering of volatility—elements that are essential in markets. GARCH-type frameworks for volatility modelling enable the capture of time-varying variance, asymmetric effects, and shock persistence, which are prevalent in stock returns.

An essential factor in this modelling framework is the determination of the suitable number of lags for the autoregressive (AR) element of the mean equation. Utilising an insufficient number

of lags undermines the inclusion of crucial historical return dependencies, whilst an excessive quantity of lags might lead to overfitting and unstable models.

This study employs the Akaike Information Criterion (AIC) as the criterion for model selection. AIC balances model fit and complexity by imposing penalties on models with an increased number of parameters. The lag order with the minimal AIC value is chosen as it signifies the optimal balance between explanatory efficacy and simplicity.

The application of the AIC across a spectrum of delays, generally from 1 to 5, is a recognised and an effective method in time series analysis since it reconciles model fit with complexity. This range adequately captures the short-term autocorrelations prevalent in index returns, without introducing extraneous noise or overfitting. It guarantees computational efficiency and conforms to established procedures in empirical studies. By contrasting models with varying lag durations and choosing the one with the minimal AIC value, it is reliable to ascertain a pragmatic and well-fitting model. The application of AIC from lags 1 to 5 constitutes a statistically sound and efficient approach for lag selection.

An ARCH effect test is conducted on the residuals of the AR model to assess the appropriateness of GARCH-type modelling. This test assesses if the variance of the error terms demonstrates conditional heteroskedasticity, a crucial need for the use of GARCH models. A notable outcome from this test indicates the existence of volatility clustering, wherein substantial fluctuations in returns are likely to be succeeded by more major fluctuations, hence validating the shift to conditional variance modelling.

After this, we move on to model stock return volatility using the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model developed by Bollerslev (1986), along with its asymmetric extensions—Exponential GARCH (EGARCH) and Threshold GARCH (TGARCH).

3.2.3.1 Generalized Autoregressive Conditional Heteroskedasticity (GARCH (1,1)) Model:

The standard GARCH (1,1) model assumes that the variance of the error term is not constant over time but depends on past squared residuals (ARCH term) and past conditional variances (GARCH term). The GARCH (1,1) specification is given by:

Mean Equation:

$$R_t = \alpha_0 + \alpha_1 D_{1t} + \alpha_2 D_{2t} + \alpha_3 D_{3t} + \alpha_4 D_{4t} + \phi R_{t-1} + \varepsilon_t \quad (6)$$

Variance Equation:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (7)$$

In Equation (6), the mean equation, R_t represents the return on day t , while D_{1t} to D_{4t} are dummy variables for Monday through Thursday. Coefficients α_1 to α_4 capture the weekday return differences relative to Friday. ϕR_{t-1} accounts for autocorrelation by incorporating the previous day's return, and ε_t is the error term.

In Equation (7), the variance equation, σ_t^2 denotes the conditional variance of returns. ω is a constant, $\alpha \varepsilon_{t-1}^2$ represents the impact of past return shocks (ARCH effect), and $\beta \sigma_{t-1}^2$ captures the persistence of volatility over time (GARCH effect).

This regression model incorporates weekday dummy variables in both the mean and potentially the variance equation, facilitating the identification of day-of-the-week influences on return levels and volatility. This model aids in recognising the effects of information and shocks on market volatility over time. A significant value of " $\alpha + \beta$ " approaching to "1" indicates enduring volatility.

3.2.3.2 Exponential GARCH (EGARCH (1,1)) Model:

The EGARCH model introduced by Nelson (1991) is utilised to address the asymmetric influence of shocks, particularly the leverage effect, wherein negative shocks (negative news) exert a greater effect on volatility than positive shocks of same magnitude. EGARCH is used to detect asymmetric volatility responses, enabling the model to assess whether bad news disproportionately drives market risk—critical for understanding investor sentiment shifts across weekdays during COVID-19. The formula for conditional variance in the EGARCH model is shown below (Equation (8)).

$$\log(\sigma_t^2) = \omega + \beta \log(\sigma_{t-1}^2) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \alpha \left(\left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| - E \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| \right) \quad (8)$$

In Equation (8), $\log(\sigma_t^2)$ is the natural logarithm of the conditional variance, ensuring positivity of variance without parameter constraints. ω is a constant term, and $\beta \log(\sigma_{t-1}^2)$

captures the persistence of past volatility. The term $\frac{\varepsilon_{t-1}}{\sigma_{t-1}}$ is the standardized shock, with γ measuring the asymmetric impact of shocks—capturing the leverage effect, where negative shocks have a different impact on volatility than positive ones. The α coefficient accounts for the magnitude of shocks, adjusted by the expected value $E \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right|$, making the model responsive to unexpected large movements in returns.

EGARCH is especially useful when bad news (negative returns) is found to increase volatility more than good news of the same size, also termed as the leverage effect. A significant and negative γ would suggest the presence of asymmetry, strengthening the argument that Indian markets do not fully align with EMH, as volatility responds differently based on the direction of shocks.

3.2.3.3 Threshold GARCH (TGARCH (1,1)) Model:

The TGARCH model, proposed by Glosten, Jagannathan, and Runkle (1993), introduces a dummy variable to separately capture the effect of negative shocks on volatility. TGARCH helps reveal if downside risk is more impactful than upside movements across the week, offering behavioural insights into how fear or uncertainty amplifies volatility in emerging markets like India.

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I_{t-1} + \beta \sigma_{t-1}^2 \quad (9)$$

The variables in equation (9) are similar to those of GARCH (1,1) mentioned in equation (6), however, The key feature is $\gamma \varepsilon_{t-1}^2 I_{t-1}$, where I_{t-1} is an indicator function that equals “1” if the past shock ε_{t-1} is negative, and “0” otherwise.

Where, $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$, and 0 otherwise. Here, $\gamma \gg 0$ indicates the presence of leverage effects.

Overall, GARCH-based modelling methodology fulfils two fundamental objectives in this research:

- **Modelling Time-Varying Volatility:** Volatility is not uniform across days; it exhibits clustering and fluctuations over time, especially during crisis periods like COVID-19. GARCH models encapsulate this characteristic, enabling the examination of return volatility fluctuations throughout the weekdays.

- Evaluating EMH via Volatility: The Efficient Market Hypothesis (EMH) posits that stock prices should exhibit a random walk with erratic variance. Nonetheless, pronounced weekday patterns in volatility or asymmetric reactions to shocks indicate market inefficiencies—an essential aspect of this study.

To investigate volatility and weekday effects using GARCH models, the following hypotheses are tested:

- **H₀:** There are no day-of-the-week effects in conditional return volatility.
- **H₁:** There exist significant weekday effects in conditional return volatility.

In asymmetric models (EGARCH, TGARCH), we additionally test:

- **H₀:** Positive and negative shocks have symmetric impacts on volatility.
- **H₁:** Negative shocks have greater effects (leverage effect).

According to prior literature, it is anticipated that:

Volatility will be elevated in the beginning of the week (e.g., Mondays, due to the accumulation of information over the weekend).

Asymmetry will exist, with adverse news exerting a more pronounced impact on volatility. GARCH terms exhibit significant persistence, signifying prolonged memory in volatility processes.

3.2.4 Volume and Illiquidity Analysis:

The fourth element of this research seeks to evaluate the influence of market liquidity and trading volume on the day-of-the-week effects in Indian stock markets. This expansion directly utilises the paradigm established by Amihud (2002) and referenced in contemporary literature, such as Khan et al. (2021), indicating that less liquidity and heightened trading frictions may intensify or induce calendar anomalies.

The primary measure of liquidity employed in this study is the Amihud illiquidity ratio, which captures the price impact per unit of trading volume. Mathematically, it is calculated as:

$$ILLIQ_t = \frac{|R_t|}{V_t} \quad (10)$$

In Equation (10), R_t represents the absolute value of the logarithmic return on day t , whereas V_t denotes the volume traded on that day. This ratio is computed daily for both indices (NSE's

Nifty 50 and BSE's Sensex). The justification for employing this indicator is its advantage in indicating the ease or difficulty of trading equities without influencing the price variable. A high value of Amihud ratio signifies diminished liquidity and increased price sensitivity to transactions, alternatively, a lower value of this ratio implies high liquidity and decreased price sensitivity to transactions.

To mitigate simultaneity bias in the regression analysis, the illiquidity measure is lagged by one day (i.e., $ILLIQ_{t-1}$) to accurately represent the liquidity conditions preceding the return under examination. This lag structure posits that historical market frictions may affect investor behaviour and market results on the following day.

Equation (11) shows the regression model used for illiquidity analysis:

$$R_t = \beta_0 + \beta_1 \text{Mon}_t + \beta_2 \text{Tue}_t + \beta_3 \text{Wed}_t + \beta_4 \text{Thu}_t + \beta_5 \text{ILLIQ}_{t-1} + \sum_{i=1}^4 \delta_i (\text{DOW}_i \times \text{ILLIQ}_{t-1}) + \epsilon_t \quad (11)$$

Where:

- R_t is the daily return,
- $\text{Mon}_t, \text{Tue}_t, \text{Wed}_t$, and Thu_t are dummy variables,
- ILLIQ_{t-1} is the lagged illiquidity,
- DOW_i represents each weekday dummy ($i = 1$ to 4 for Monday to Thursday)
- $\text{DOW}_i \times \text{ILLIQ}_{t-1}$ represents the interaction terms,
- ϵ_t is the error term.

Before evaluating the illiquidity model, the study examined the illiquidity measure using summary statistics and box plot visualisations to identify extreme values. There was significant presence of outliers, and hence, this variable was winsorized at the 5th and 95th percentiles. Such an approach helped reduce the influence of extreme values while preserving the overall distribution of the variable, ensuring more robust regression results.

Furthermore, this procedure also maintained the usual structured process for testing time series regressions, i.e. by following checks for stationarity by making use of ADF tests, performing BG test for autocorrelation and later correcting for it by making use of Newey west procedure which were mentioned in the OLS approach section previously.

The key objective of this model is to identify whether illiquidity has a statistically significant influence on daily returns, both in general and conditional on the day of the week. A significant coefficient for the lagged illiquidity term would indicate that liquidity has a direct impact on next-day returns. Significant coefficients on the interaction terms would suggest that the impact of illiquidity varies across the week, thereby offering a liquidity-based explanation for the observed day-of-the-week effects.

3.2.5 Efficient Market Hypothesis (EMH) Testing:

The last part of this methodology is testing the Efficient Market Hypothesis (EMH), especially the weak-form efficiency, which says that current stock prices fully reflect all information about past prices. If the market is weak-form efficient, then prices or returns from the past shouldn't be able to tell anything about returns in the future. So, any proof of correlations in return series or the ability to predict future values based on past values would go against this idea of market efficiency.

This study adopts an Autoregressive modelling (AR) approach. Autoregressive modelling regresses the current daily log returns on their own lagged values over a specified number of previous periods.

If the coefficients of the lagged return variables are significantly different from zero, it means that past returns can help predict future returns, which suggests that weak-form EMH might not hold true.

Equation (12) shows the regression model used for testing EMH:

$$R_t = \alpha + \sum_{i=1}^p \beta_i R_{t-i} + \varepsilon_t \quad (12)$$

Where:

- R_t is the return at time t
- R_{t-i} is the return lagged by i days
- α is the constant (intercept)
- β_i is the coefficient for the i -th lag
- ε_t is the white noise error term

It is crucial to confirm the return series is stationary before calculating the AR model. Non-stationary series could lead to erroneous conclusions such as false test statistics and higher R-squared values. The Augmented Dickey-Fuller (ADF) test is used to test the null hypothesis that a unit root exists in the return series, or non-stationarity. Rejection of the null hypothesis verifies that the returns are stationary and appropriate for time-series analysis.

After stationarity is verified, autoregressive models with rising lag order are evaluated to find the most suitable lag structure for every index (Sensex and Nifty 50). This model also systematically compares the fit of AR models with 1 to 5 lags using an Akaike Information Criterion (AIC)-based method which was explained in the volatility modelling section previously. The lowest AIC model is picked as the best specification, guaranteeing that the selected lag duration has the most explanatory power without needlessly raising the parameter count.

To assess the presence of return predictability and test the weak-form Efficient Market Hypothesis (EMH), AR (5) models were fitted to both indices, with the optimal lag structure selected using the Akaike Information Criterion (AIC) as outlined earlier. The significance of lagged return coefficients was tested using t-statistics to determine whether past returns could predict current returns. A significant lagged coefficient means that past returns can help predict future returns, which goes against weak-form efficiency. The overall model fit is further evaluated using R-squared and the F-statistic, where a significant F-statistic suggests that the group of lagged terms explains a meaningful portion of return variation. While low R-squared values are typical for financial time series due to their stochastic nature, they do not imply model failure. Lastly, the Breusch-Godfrey test was used to detect for residual autocorrelation, ensuring the model adequately captured temporal dependencies. These procedures—as detailed in Section 3.2.5 of the methodology—help validate the robustness and reliability of the EMH testing results presented in the analysis.

This methodological framework complements comparable strategies used in earlier academic studies on market efficiency and is in line with accepted econometric procedures. Eventually, this part of the research intends to evaluate if past return data might account for future stock movements in the Nifty 50 and Sensex indices, which would strengthen insights on the degree of informational efficiency in India's equity markets.

While the methodology adopted in this study adheres to robust time-series econometric practices, several limitations warrant consideration. First, potential multicollinearity among weekday dummies and their interactions may reduce the precision of estimated coefficients, especially in models with interaction terms (Gujarati & Porter, 2009). Second, the exclusion of macroeconomic control variables such as interest rates, inflation, or geopolitical developments raises concerns about omitted variable bias, which can result in biased and inconsistent estimates (Wooldridge, 2012). Third, despite tests for stationarity and autocorrelation, the analysis remains susceptible to data mining and spurious correlations—a well-documented issue in calendar anomaly research, particularly under repeated hypothesis testing (Sullivan, Timmermann & White, 2001; Schwert, 2003). Fourth, regarding processes of volatility modelling, reliance solely on AIC for model selection could be insufficient in highly volatile markets; incorporating alternative criteria like Bayesian Information Criterion or conducting likelihood ratio tests could enhance robustness (Burnham & Anderson, 2002). Moreover, the study does not explicitly test for volatility spillovers or parameter stability, both of which are critical in capturing the full dynamics of volatility during periods such as the COVID-19 crisis (Engle, 2004; Bollerslev et al., 1992). Lastly, while the Amihud illiquidity measure is lagged to mitigate simultaneity bias (i.e., the risk that returns and liquidity may influence each other simultaneously, making causal direction unclear), it may not fully capture delayed or nonlinear liquidity effects in emerging markets (Amihud, 2002). These constraints highlight the need for further analysis of results which could offer directions for future methodological refinement.

4. Results:

This section describes all the results obtained from the methodology process followed in the above section applied on the Nifty50 and Sensex indices.

4.1 OLS Approach Results:

The OLS regressions were estimated on the full sample for both the BSE Sensex and NSE Nifty50 indices using weekday dummy variables. The purpose was to determine whether systematic differences in mean returns exist across trading days, a violation of weak-form efficiency. The lagged return modified regression model was induced to control for autocorrelation, and Newey West standard errors were applied for robust inference.

From Table 4, BSE's full-sample model, only Tuesday returns were statistically significant and positive ($\beta = 0.0023$, $p = 0.036$), suggesting a strong weekday anomaly. All other weekday

coefficients were statistically insignificant, furthermore, lagged return variable showed no meaningful predictive power too. The joint F-test for all weekday dummies was significant (F-statistic = 3.07, $p = 0.0158$), affirming that at least one of the weekdays returns differed from Friday. Although the model's R^2 was low (0.0113), this is typical of high-frequency financial return models, reflecting the stochastic nature of stock prices.

In contrast, the results for NSE full-sample model from table 4 reveals a significant Wednesday effect ($\beta = 0.0024$, $p = 0.005$), while Tuesday and Thursday remained insignificant. The lagged return was not statistically significant ($p = 0.125$). Notably, the joint F-test was significant ($F = 3.37$, $p = 0.0050$), confirming the presence of weekday-based return differentials. This model's explanatory power also remained low ($R^2 = 0.0135$), but the statistical significance of Wednesday suggests systematic market behaviour on that day.

Table 4: Shows the full sample and lagged OLS models for both Sensex and Nifty50 indices.

Regressions:	(i)	(ii)	(iii)	(iv)
	BSE (SENSEX) full sample	BSE (SENSEX) lagged returns	NSE (NIFTY50) full sample	NSE (NIFTY50) lagged returns
Dependent Variables	return	return	return	return
Lagged return	-	-0.0442 [-1.56]	-	-0.0436 [-1.54]
Monday	-0.0018 [-1.62]	-0.0017 [-1.57]	0.0035 [0.29]	0.0035 [0.28]
Tuesday	0.0023** [2.08]	0.0023** [2.06]	-0.0017*** [-1.81]	-0.0017*** [-1.78]
Wednesday	0.0003 [0.29]	0.0004 [0.40]	0.0024** [2.53]	0.0023** [2.47]
Thursday	-0.0002 [-0.22]	-0.0001 [-0.16]	0.0004 [0.43]	0.0005 [0.52]
Constant	0.0003 [0.45]	0.0003 [0.42]	0.0002 [0.40]	0.0002 [0.42]
No. of observations	1237	1236	1238	1237
R²	0.0113	0.0133	0.0116	0.0135

t-statistics values are depicted in [], * represents 1% significance level, **=5% significance level and ***=10% significance level.

Table 5 presents the OLS regressions with lagged returns for the BSE Sensex across pre-COVID, COVID, and post-COVID phases. Across all three sub-periods, Tuesday returns remain statistically significant at the 5% level ($\beta = 0.0023$, $t = 2.08$), highlighting a persistent weekday anomaly. This recurring Tuesday effect is consistent and stable throughout the three phases, suggesting a robust behavioural or institutional source. No other weekday shows statistical significance in any sub-period. The dummy variables for pre-COVID, COVID, and post-COVID periods themselves are insignificant, indicating that baseline returns outside weekday effects remained stable. The lagged return term is also insignificant across all models, supporting the idea that past return information does not predict current returns—consistent with weak-form EMH for the Sensex.

Table 5: Shows the lagged OLS models for the Sensex index for three regressions (from i-iii) pre-covid, during covid and post-covid periods respectively.

regressions	(i)	(ii)	(iii)
	return	return	return
L1 return	-0.0443 [-1.04]	-0.0443 [-1.04]	-0.0443 [-1.03]
Monday	-0.0018 [-1.33]	-0.0018 [-1.32]	-0.0018 [-1.32]
Tuesday	0.0023** [2.08]	0.0023** [2.08]	0.0023** [2.08]
Wednesday	0.0005 [0.45]	0.0005 [0.46]	0.0005 [0.46]
Thursday	-0.0002 [-0.16]	-0.0002 [-0.16]	-0.0002 [-0.16]
pre_covid	-0.0001 [-0.17]	-	-
covid	-	0.0001 [0.08]	-
post_covid	-	-	0 [0.06]
Constant	0.0004 [0.50]	0.0004 [0.38]	0.0003 [0.42]
No. of observations	1236	1236	1236

t-statistics values are depicted in [], * represents 1% significance level, **=5% significance level and ***=10% significance level.

Table 6 shows the similar lagged OLS regressions for Nifty50. In contrast to Sensex, this index exhibits a statistically significant Monday effect ($\beta = 0.0036$, $t > 5.8$) and Wednesday effect ($\beta = 0.0024$, $t = 2.74$) consistently across all three sub-periods. Unlike Sensex, no significant Tuesday effect is detected. This contrast reveals that Nifty50 exhibits stronger early-week return anomalies, while Sensex's anomalies are mid-week specific (Tuesday).

Table 6: Shows the lagged OLS models for the Nifty50 index for three regressions (regression a, b and c) pre-covid, during covid and post-covid periods respectively.

regressions	(a)	(b)	(c)
	return	return	return
L1 return	-0.0438 [-1]	-0.0437 [-0.99]	-0.0437 [-0.99]
Monday	0.0036* [6.3]	0.0035* [6.89]	0.0036* [5.81]
Tuesday	-0.0017 [-1.46]	-0.0017 [-1.45]	-0.0017 [-1.46]
Wednesday	0.0024* [2.74]	0.0024* [2.74]	0.0024* [2.74]
Thursday	0.0005 [0.66]	0.0005 [0.66]	0.0005 [0.67]
pre_covid	-0.0002 [-0.34]	-	-
covid	-	0.0001 [0.08]	-
post_covid	-	-	0.0002 [0.22]
Constant	0.0003 [0.52]	0.0002 [0.41]	0.0002 [0.28]
No. of observations	1237	1237	1237

t-statistics values are depicted in [], * represents 1% significance level, **=5% significance level and ***=10% significance level.

The differences seen in Table 5 and Table 6 may stem from index composition: Nifty50 includes more diverse sectors and higher-frequency traders, potentially contributing to a

stronger Monday effect due to institutional rebalancing. In contrast, Sensex may reflect more traditional market reactions tied to delayed institutional flows. The persistence of these effects may suggest that behavioural biases or information processing cycles that are distinct across each index.

In summary, weekday effects are evident and vary by index and timing. NSE shows stronger anomalies on Monday and Wednesday, while BSE consistently highlights Tuesday effects. These anomalies could reflect partial market inefficiencies likely exacerbated by trading behaviours, institutional flows, or post-weekend investor sentiment.

4.2 Structural Break Testing Results:

To examine whether the COVID-19 pandemic introduced a structural shift in day-of-the-week return patterns, interaction regressions were conducted using a structural break dummy set on 31 January 2020. This dummy was interacted with weekday dummies to test for regime-specific shifts in return behaviour. Later, Chow test framework was applied to evaluate the joint significance of the interaction terms.

4.2.1 BSE Structural Break Analysis (SENSEX)

From Table 7, the interaction term `tue_covid` for BSE is statistically significant ($\beta = 0.0053$, $t = 2.35$), indicating a notable increase in Tuesday returns following the COVID-19 breakpoint. This suggests a structural change in investor behaviour on Tuesdays, possibly due to intensified response to weekend news or institutional positioning early in the week. All other weekday interaction terms (`mon_covid`, `wed_covid`, `thu_covid`) are statistically insignificant, pointing to limited shifts in weekday behaviour beyond Tuesday. The Chow test for BSE, reported in Table 8, yields an F-statistic of 2.66 ($p = 0.0213$), leading to the rejection of the null hypothesis of structural stability. This confirms a statistically significant structural break in return patterns across weekdays post-COVID. These findings suggest that although the Sensex largely retained its weekday return structure, a distinct shift occurred on Tuesdays, aligning with prior results from OLS regressions.

4.2.2 NSE Structural Break Analysis (NIFTY50)

In the Nifty50 regression (Table 7), a significant positive Tuesday $\times$ COVID interaction ($\beta = 0.0054$, $t = 2.43$) is observed, confirming that Tuesday returns increased following the

pandemic onset. Other interaction terms (mon_covid, wed_covid, thu_covid) are statistically insignificant, reinforcing the conclusion that the structural break was concentrated on Tuesdays. The Chow test result from Table 8 for NSE further supports this interpretation, with an F-statistic of 2.59 ($p = 0.0243$). This indicates a statistically significant structural break in weekday return patterns, specifically post-COVID. The results are consistent with those observed for the BSE, though the effect appears slightly more robust for NSE. This may reflect the NSE's broader investor base and quicker market reaction to post-weekend news or economic signals during the pandemic.

Table 7: Structural break analysis regressions for Sensex and Nifty50 indices.

Regression	(i)	(ii)
	BSE (SENSEX)	NSE (NIFTY50)
Dependent Variables	return	return
Monday	-0.0001 [-0.11]	-0.0005 [-0.32]
Tuesday	-0.0007** [-0.46]	-0.0009** [-0.55]
Wednesday	-0.0005 [-0.33]	-0.0008 [-0.50]
Thursday	-0.0008 [-0.48]	-0.0011 [-0.60]
COVID	-0.0008 [-0.53]	-0.0009 [-0.59]
mon_covid	-0.0027 [-1.21]	-0.0023 [-1.03]
tue_covid	0.0053 [2.35]	0.0054 [2.43]
wed_covid	0.0015 [0.68]	0.0018 [0.82]
thu_covid	0.0011 [0.44]	0.0011 [0.51]
Constant	0.0008 [0.70]	0.0009 [0.79]
No. of observations	1237	1238
R ²	0.0219	0.0219

t-statistics values are depicted in [], **=5% significance level.

Table 8: Chow test outputs for Sensex and Nifty50 indices.

No.	Chow Test	Variables tested	Test statistics	p-value	Conclusion
1	Sensex returns	COVID, mon_covid, tue_covid, wed_covid, thu_covid	F=2.66	0.0213	Structural break is present
2	NSE (NIFTY50)	COVID, mon_covid, tue_covid, wed_covid, thu_covid	F=2.59	0.0243	Structural break is present

4.3 Volatility Modelling:

The results obtained from the volatility modelling for both the indices are explained here:

4.3.1 Volatility Model Results for Bombay Stock Exchange:

Table 9 presents the GARCH-family regression estimates for BSE. The GARCH (1,1) model indicates a statistically significant Tuesday return ($p = 0.016$), providing strong evidence against the null hypothesis of no weekday effect for Tuesday. This also holds economic significance: the positive Tuesday return ($\beta = 0.0019$) implies a notable weekday-based return premium, particularly relevant for short-term trading strategies. The lagged return variable was not significant, supporting the null hypothesis under weak-form EMH that past returns do not predict future values. The sum of the ARCH and GARCH terms is $\alpha + \beta \approx 0.978$, close to 1, confirming strong volatility persistence and validating the alternative hypothesis of conditional heteroskedasticity. The GARCH framework is therefore appropriate for modelling return variance in BSE.

The EGARCH (1,1) model captures asymmetric volatility. Though weekday dummies are not significant at the 5% level, Tuesday approaches marginal significance ($p = 0.092$), consistent with earlier findings. The EGARCH coefficient is negative and significant, supporting the alternative hypothesis of a leverage effect—that negative shocks induce greater volatility than positive ones. This reflects a key behavioural feature where markets respond more sharply to bad news. The ARCH term is also significant, indicating that recent shocks substantially influence current volatility.

The TGARCH (1,1) model further confirms the asymmetry in volatility. Tuesday returns are again statistically significant, and the lagged return is also significant, suggesting short-term

momentum and contradicting weak-form efficiency for this specification. The TGARCH term is statistically and economically significant, confirming the presence of leverage effects. Together, these results reject the null hypothesis of symmetric volatility and reinforce that BSE exhibits both weekday effects and asymmetric volatility, particularly around Tuesday.

Table 9: volatility model outputs for Sensex index.

Variable	GARCH (1,1)	EGARCH (1,1)	TGARCH (1,1)
mon	0.0002 [-0.31]	-0.0016 [-1.45]	0.0002 [-0.45]
tue	0.0019** [-2.42]	0.0019* [-1.69]	0.0019** [-2.58]
wed	-0.0001 [-0.05]	0.0002 [-0.23]	0.0004 [-0.52]
thu	0.0005 [-0.8]	-0.0004 [-0.35]	0.0007 [-1.04]
L1.return	0.045 [-1.47]	-0.0237 [-1.36]	0.072** [-2.65]
const	0.0003 [-0.67]	0.0006 [-0.79]	-0.0002 [-0.52]
ARCH (1)	0.126*** [-8.39]	62.21*** [-29.61]	0.196*** [-9.22]
GARCH (1)	0.852*** [-46.26]	-	0.883*** [-73.01]
EGARCH (1)	-	-1.01*** [-819.91]	-
TGARCH (1)	-	-	-0.216*** [-9.18]

t-statistics values are depicted in [], * represents 1% significance level, **=5% significance level and ***=10% significance level.

4.3.2 Volatility Model Results for National Stock Exchange:

From Table 10, Tuesday coefficient is statistically significant at the 5% level in the GARCH (1,1) model ($\beta = 0.00185$, $t = 2.24$), suggesting a significant Tuesday effect on returns. This implies that the average return on Tuesdays is higher than that on the reference day (Friday). There is no obvious evidence of return differentials on the other weekday variables (Monday, Wednesday, Thursday), as they are statistically insignificant. The data confirms strong

volatility clustering, as evidenced by the volatility dynamics, which are depicted through significant ARCH ($\alpha = 0.129$) and GARCH ($\beta = 0.846$) terms.

The Tuesday effect remains statistically significant in the EGARCH (1,1) model ($\beta = 0.00194$, $t = 2.02$), which supports the previous discovery. Furthermore, the ARCH (0.854) and EGARCH (298.83) terms are highly significant, demonstrating that asymmetric volatility effects are captured. This suggests that negative shocks have a distinct impact on volatility than positive ones. The leverage effect is consistent with the economic implication that the market reacts more firmly to bad news.

In contrast, the TGARCH (1,1) model also recognises Tuesday as statistically significant ($\beta = 0.00181$, $t = 2.34$), and it also captures asymmetric volatility through the significant TARCH term (-0.216 , $t = -9.01$). This further implies the presence of nonlinearities in the behaviour of return volatility.

Table 10: volatility model outputs for Nifty50 index.

Variable	GARCH (1,1)	EGARCH (1,1)	TGARCH (1,1)
mon	0.000005 [-0.01]	-0.00037 [-0.47]	0.00007 [-0.11]
tue	0.00185** [-2.24]	0.00194** [-2.02]	0.00181** [-2.34]
wed	-0.00021 [-0.25]	0.00004 [-0.04]	0.00022 [-0.27]
thu	0.00039 [-0.52]	-0.00022 [-0.25]	0.00056 [-0.74]
L1.return	0.049 [-1.59]	0.026 [-1]	0.079*** [-2.91]
const	0.00041 [-0.77]	0.00037 [-0.62]	-0.00017 [-0.33]
ARCH(1)	0.129*** [-8.14]	298.83*** [-12.11]	0.197*** [-9.15]
GARCH(1)	0.846*** [-43.49]	-	0.880*** [-72.16]
EGARCH(1)	-	0.854*** [-67.05]	-
TGARCH(1)	-	-	-0.216*** [-9.01]

t-statistics values are depicted in [], * represents 1% significance level, **=5% significance level and ***=10% significance level.

The null hypothesis of no Tuesday effect is consistently rejected, as confirmed by hypothesis testing, while it is retained for other days. The weak-form EMH is challenged by the lagged return in TGARCH. The Tuesday effect's consistency and the robustness of volatility clustering across models are critical indicators of partial inefficiency and behavioural biases, which may be associated with investor sentiment and institutional trading patterns. The significance of employing sophisticated volatility models in emergent markets such as India is emphasised by these patterns.

4.4 Volume and Illiquidity Analysis Results:

This section examines whether market illiquidity can explain the observed day-of-the-week effects in returns by incorporating the Amihud Illiquidity Ratio and its interaction with weekday dummies. Table 11 displays the regression results for both the indices in study, where the lagged illiquidity measure and its weekday interactions were included as explanatory variables in the return equations.

Starting with the BSE's results in Table 11, the coefficient for Monday $\times$ Illiquidity is statistically significant ($p = 0.001$), with a negative sign ($\beta = -8,174,737$). This indicates that higher illiquidity on Mondays leads to lower returns. No other weekday-illiquidity interaction terms are statistically significant at the 5% level, and the lagged illiquidity variable itself is not significant. The significant Monday interaction suggests that on days following the weekend—when trading volumes may be thin and information lags could occur—investors are less willing to trade, leading to a reduction in prices and returns.

For the NSE's Nifty50 results from Table 11, the results reveal a broader relationship between illiquidity and weekday returns. Both Monday $\times$ Illiquidity ($\beta = -5,874,963$, $p = 0.012$) and Tuesday $\times$ Illiquidity ($\beta = 6,514,662$, $p = 0.002$) interactions are statistically significant. The negative Monday coefficient, like BSE, confirms the presence of a liquidity-related return drop at the start of the week. However, the positive and highly significant Tuesday interaction suggests a rebound or reversal effect, where markets recover from Monday's low-volume pressure and post positive returns on Tuesday in response to accumulated information or liquidity adjustments.

Interestingly, the standalone Tuesday dummy for NSE is also marginally significant ($p = 0.081$), further reinforcing the argument that Tuesday's experience stronger return activity, possibly due to a delay in trade execution or reaction to new information carried over from the weekend and Monday. For BSE, however, none of the weekday dummies are independently

significant when illiquidity is accounted for, implying that illiquidity plays a stronger role in shaping Sensex's weekday return dynamics only through the Monday channel.

The contrasting results between the two indices suggest that the impact of illiquidity on return behaviour is more prominent in NSE, where trading activity is broader, and more retail driven. This makes the market more sensitive to volume-related pressures and investor sentiment shifts early in the week. In contrast, BSE's narrower and more institutionally focused trading base might reflect more suppressed reactions to illiquidity except for the critical Monday adjustment.

Table 11: Volume and Illiquidity model regression outputs for Sensex and Nifty50 index.

	SENSEX	NIFTY50
Variable	return	return
Monday	0.002631 [1.49]	-0.0019*** [-1.78]
Tuesday	0.001123 [0.61]	0.002*** [1.81]
Wednesday	0.00236 [1.32]	0.0002 [0.2]
Thursday	0.000324 [0.18]	-0.0003 [-0.3]
Lagged Illiq	2988739 [1.64]	3381599 [-0.3]
Mon * Illiq	-81,74,737*** [-3.23]	-8147323** [-2.51]
Tue * Illiq	14,20,977 [0.57]	908157** [3.17]
Wed * Illiq	-38,67,277 [-1.45]	2788618 [0.08]
Thu * Illiq	-9,65,008 [-0.36]	-957011 [-0.32]
Constant	-0.001242 [-0.99]	0.0003 [0.72]

t-statistics values are depicted in [], * represents 1% significance level, **=5% significance level and ***=10% significance level.

In terms of hypothesis testing, the results allow us to reject the null hypothesis that illiquidity has no effect on weekday returns, especially for Monday (both indices) and Tuesday (NSE). These outcomes support the alternative hypothesis that market frictions such as illiquidity partially drive day-of-the-week return anomalies in the Indian stock market.

Volume and liquidity effects offer a plausible explanation for the day-of-the-week anomaly in a more comprehensive context. Lower liquidity on Mondays is likely to result in return suppression for both the BSE and NSE, because of reduced trading activity or cautious investor behaviour following the weekend. This is further supported by the statistically significant negative coefficients for Monday-illiquidity in both indices. The significance of Tuesday returns implies a type of compensatory trading behaviour, in which participants react to the inefficiencies of Monday. Although the BSE results exhibit comparable patterns, the significance levels for the NSE are more robust, suggesting that the NIFTY50 index is more susceptible to liquidity fluctuations. These findings substantiate the notion that weekday return fluctuations are influenced by market microstructure factors, such as liquidity, rather than solely by calendar anomalies. In emerging markets, where structural and behavioural frictions are more prevalent, our comprehension of market efficiency and investor behaviour across trading days is enhanced by the recognition of volume effects.

4.5 Efficient Market Hypothesis (EMH) Testing Results:

Table 12: Output of Autoregressive (5) model for Sensex.

Variable	Coefficient	Std. Err	z-stat	p-value
L1.return	0.0571	0.0176	3.24	0.0012
Constant	0.0005	0.0004	1.23	0.217

Table 13: Output of Autoregressive (5) model for Nifty50.

Variable	Coefficient	Std. Err	z-stat	p-value
L1.return	0.0479	0.0182	2.64	0.0082
Constant	0.0004	0.0004	1.16	0.247

The regression results for NSE, shown in Table 13, indicate that the optimal lag length was selected using the AIC, with the best fit found at AR (5). The lag 5 (L5) coefficient is statistically significant at the 1% level ($\beta = 0.0479$, $p = 0.0082$), suggesting a small but meaningful autocorrelation in returns. This finding contradicts the weak-form Efficient Market Hypothesis (EMH), which holds that past returns should not predict future returns.

Similarly, the best-fitting model for BSE was also AR (5), with the lag 5 coefficient significant at the 1% level ($p = 0.0012$). This again points to serial dependence in returns and provides further evidence against the weak-form EMH for the Indian market.

From Table 12 and Table 13, The lag 5 coefficient for Sensex is 0.0571 ($z = 3.24$, $p = 0.0012$), and for Nifty50, it is 0.0479 ($z = 2.64$, $p = 0.0082$). While these coefficients are modest in size, their statistical significance indicates that returns are not entirely random, and that some degree of short-term predictability exists in both markets.

Although the magnitude of the coefficients is small—around 5%—they still provide evidence of weak autocorrelation in daily returns. This suggests that technical traders could potentially exploit short-term patterns, though the limited size of this effect means any real-world gains could be minimal after accounting for transaction costs and liquidity constraints.

The significant lagged return coefficients at the 1% level for both NSE and BSE indicate that past returns have predictive power, suggesting that Indian stock markets did not fully exhibit weak-form efficiency during the sample period. This outcome is consistent with findings in other emerging markets, where slower price adjustments and information asymmetries often lead to partial inefficiencies.

Overall, while the predictive power of past returns is limited in economic terms, the consistent and statistically significant autocorrelation found in both NSE and BSE suggests that Indian markets exhibit partial inefficiencies, especially over short-term horizons.

5. Discussions:

The empirical analysis of the NSE and BSE datasets uncovered significant findings about calendar anomalies, specifically the day-of-the-week effects, in Indian stock markets throughout the five-year period from January 2018 to January 2023. The study employed a multifaceted econometric approach, incorporating OLS regressions, autoregressive models, GARCH-family models, structural break analysis, and illiquidity interaction models to analyse market behaviour over different time periods and economic volatility regimes.

Ordinary Least Squares regressions indicated slight yet discernible variability in daily returns. Tuesday exhibited persistent significance for NSE (Nifty 50) across many models, particularly during structural break regressions, suggesting a calendar abnormality on this day. Comparable tendencies were observed in the BSE (Sensex), albeit with slightly diminished persistence. Autoregressive lag models identified modest yet statistically significant autocorrelation in historical returns, so partially contesting the weak-form Efficient Market Hypothesis (EMH). Structural break models identifying COVID-19 as the breakpoint demonstrated substantial alterations in weekday coefficients, particularly a notable Tuesday effect in NSE following the pandemic.

The GARCH (1,1), EGARCH (1,1), and TGARCH (1,1) models validated the existence of time-varying volatility. The GARCH and TGARCH models exhibited statistically significant volatility clustering and asymmetric reactions to shocks. TGARCH models also demonstrated negative asymmetries that align with leverage effects. Illiquidity regressions indicated that for BSE, lagged Amihud illiquidity measurements, particularly on Mondays, exhibited a statistically significant negative correlation with returns, implying liquidity-driven price anomalies.

The EMH regressions revealed substantial autoregressive coefficients in the lag terms, hence challenging the concept of random walk behaviour in Indian marketplaces. The findings reveal intricate evidence of inefficiencies, behavioural tendencies, and liquidity-induced abnormalities, particularly in the context of COVID-19 disruptions. The findings confirm the presence of day-of-the-week influences on both the NSE and BSE. The Tuesday effect is the most consistently substantial across both indices, especially in the post-COVID structural break study. Although previous days had intermittent significance, Tuesday demonstrated statistical consistency, hence underscoring a possible anomaly. These data substantiate the notion that Indian equity markets do not entirely adhere to weak-form efficiency.

The day-of-the-week effects occurred more frequently in the NSE than in the BSE. In the NSE, Tuesday consistently exhibited substantial coefficients across OLS, structural break, and volatility models. Simultaneously, BSE exhibited diminished although persistent irregularities, predominantly occurring on Tuesdays and sporadically on Thursdays. The duration of these effects fluctuated over time, becoming more pronounced during the COVID-19 period, indicating behavioural changes or macro-level structural disturbances affecting investor sentiment and market dynamics. Tuesday was the most noted day in both indices, indicating anomalous returns compared to the excluded baseline (Friday). This was particularly evident in the volatility models and structural break analysis. The persistence and intensity of the Tuesday effect, particularly following COVID-19, indicate possible behavioural influences such as overreactions to weekend news or systematic trading patterns at the week's onset.

While these weekday anomalies were statistically significant, it is critical to highlight that their economic magnitude is limited. The average abnormal return associated with Tuesday effects, although statistically distinguishable from zero, generally ranged from 0.18% to 0.24%. These return levels may not provide strong practical implications for most investors, especially after accounting for transaction costs, taxes, and liquidity constraints. Therefore, these patterns, while academically meaningful, must be interpreted with caution in real-world investment contexts.

Additionally, this study conducted many hypothesis tests across models and timeframes, increasing the likelihood of Type I errors—false positives that may arise purely by chance. As Sullivan et al. (2001) and Schwert (2003) have noted, extensive data mining and multiple testing can lead to overstated significance. Although this study used robust model specifications and diagnostic checks, it did not apply formal adjustments like Bonferroni or Holm corrections to control for multiple comparisons. Future research should consider implementing such adjustments to validate the robustness of observed anomalies.

The structural break analysis conducted around January 31, 2020, the date on which the World Health Organisation declared COVID-19 a global health emergency, provided profound insights into the evolution of day-of-the-week effects in the Indian markets. Weekday returns in both the NSE and BSE were generally statistically insignificant prior to the structural break, and they were more closely aligned with the weak-form EMH. For example, the NSE regression did not exhibit any robust weekday anomalies, and the BSE coefficients fluctuated without consistent statistical relevance. Nevertheless, significant changes occurred in the post-

COVID era. The Tuesday dummy for NSE experienced a substantial increase in returns compared to the pre-COVID period, with a coefficient of 0.00449 and a p-value of 0.044. This corroborates the idea that investor behaviour changed in response to increased uncertainty, which may have been the result of delayed reactions to weekend news or institutional order patterns early in the week. Although some marginal effects were observed, BSE did not exhibit a significant post-COVID significance.

These volatility patterns suggest that day-of-the-week anomalies are not merely the result of irrational investor behaviour or market frictions, but also a consequence of risk dynamics. High conditional volatility on specific weekdays may be indicative of institutional trading practices, such as news announcements that concentrate at the beginning of the week or mutual fund rebalancing. This results in predictable market fluctuations in terms of volatility, which in turn influence returns. As a result, an incomplete comprehension of market efficiency and anomalies would result from analysing the weekday effects solely through return metrics without accounting for volatility.

In summary, while the study uncovered strong statistical evidence of weekday anomalies—particularly on Tuesdays—the economic impact of these effects is modest, and multiple testing concerns suggest that a degree of caution is warranted when interpreting these findings. These insights, though limited in their application to investment strategy, provide valuable contributions to our understanding of how Indian equity markets behave during periods of systemic disruption.

6. Conclusion:

This dissertation set out to investigate the persistence and nature of the day-of-the-week (DOW) effect in Indian equity markets, using a robust econometric framework over a critical five-year period encompassing the pre-COVID, COVID, and post-COVID phases. The study focused on two of India's most representative indices—the NSE's Nifty 50 and the BSE's Sensex—and sought to identify whether weekday return anomalies persisted during times of economic uncertainty and structural change.

The research addressed three major gaps in existing literature. First, it extended the empirical assessment of DOW effects to a contemporary and highly volatile period, offering updated insights for an era dominated by digital trading, retail participation, and global market integration. Second, it integrated advanced GARCH-family models to capture the nuances of volatility asymmetry and risk clustering—dimensions often overlooked in earlier Indian studies. Third, it incorporated market microstructure elements via Amihud illiquidity analysis, enabling an examination of whether liquidity constraints contributed to weekday anomalies, an area insufficiently explored in prior Indian research.

The results obtained through OLS, AR, GARCH-type models, structural break regressions, and illiquidity analysis provided a multifaceted view of return behaviour. Both indices displayed statistically significant weekday effects at various points, with the Tuesday effect consistently emerging across multiple model specifications, particularly in the Sensex. This contrasts with previous research such as Bhattacharya et al. (2003), which found inconsistent or sector-specific weekday patterns, and more recent findings like Sahoo (2021) and Yadav & Kadian (2025), which highlighted increased anomalies during crisis periods.

The incorporation of a structural break test around January 2020—coinciding with the declaration of COVID-19 as a global health emergency—confirmed that return dynamics indeed shifted post-COVID. Both Chow tests and interaction term regressions pointed to Tuesday returns increasing significantly during the post-COVID period, especially in the NSE index. These results support the argument that systemic shocks can disrupt investor behaviour and temporarily alter return predictability.

In comparing the findings across models and indices, one observation stands out: while anomalies were found in both indices, their consistency and magnitude varied. The Nifty 50 demonstrated greater weekday sensitivity, particularly on Mondays and Wednesdays, whereas

the Sensex more prominently exhibited the Tuesday effect. This divergence could be attributed to differences in index composition, trading volume, and investor participation. The NSE, being more widely traded and institutionally driven, may be more responsive to structural changes and investor sentiment fluctuations.

The GARCH, EGARCH, and TGARCH models provided robust support for volatility clustering and asymmetry, revealing persistent conditional heteroskedasticity and evidence of leverage effects—particularly in the EGARCH and TGARCH frameworks. However, weekday dummy variables in volatility equations were not consistently significant across both indices, suggesting that while return levels varied, volatility patterns were less clearly tied to weekday behaviour. These insights refine earlier volatility-related findings by Srinivasan and Kalaivani (2013) and Aggarwal and Jha (2023) and reinforce the need for caution when generalizing weekday volatility anomalies.

A novel contribution of this study is the analysis of trading volume through Amihud's illiquidity measure. Results indicate that Monday returns in both indices were adversely impacted by heightened illiquidity, suggesting that liquidity constraints and frictions at the start of the trading week can exacerbate pricing inefficiencies. This microstructural perspective aligns with Khan et al. (2021) but adds depth by incorporating interaction terms and lag structures.

The EMH testing via autoregressive models revealed mild yet statistically significant autocorrelation at lag 5, suggesting a limited degree of return predictability—particularly during the COVID period. While these findings do not definitively refute the weak-form EMH, they highlight pockets of inefficiency, particularly during market stress. The observed autocorrelation supports the notion of delayed information assimilation, as posited in earlier anomaly literature.

Despite these significant findings, several limitations must be acknowledged. First, the use of a single structural break point simplifies what was likely a more complex and multi-phased market adjustment during COVID-19. Future work could explore multiple breakpoint models or rolling regressions to assess evolving anomaly behaviour. Second, the analysis is susceptible to false positives (Type I errors) due to multiple hypothesis testing across various models and sub-periods. This limitation, echoed by Schwert (2003) and Sullivan et al. (2001), calls for future research to apply multiple testing corrections (e.g., Bonferroni or Holm-Bonferroni procedures). Third, while this study employed lagged illiquidity terms to mitigate endogeneity,

it may not fully capture nonlinear or delayed liquidity effects, particularly in thinly traded assets or during crisis conditions.

Crucially, while the DOW effects found in this study are statistically significant, their economic significance is limited. Most weekday return premiums, such as the Tuesday effect in BSE and NSE, hovered around 0.2–0.3%, which may not offer material advantages to investors after accounting for transaction costs, bid-ask spreads, taxes, and slippage. A clear statement of caution is warranted: statistical patterns should not be misconstrued as actionable investment strategies without deeper economic validation. This is especially important in emerging markets, where frictions can erode potential gains from anomaly exploitation.

In conclusion, this dissertation has provided updated evidence on the presence and persistence of weekday return anomalies in India's equity markets under a highly turbulent global context. The research finds that day-of-the-week effects, particularly the Tuesday effect, remain observable but are not uniformly consistent across indices or time periods. Structural disruptions like COVID-19 amplify these effects temporarily, but Indian markets exhibit a gradual reversion towards efficiency. The findings support the hypothesis that while short-term inefficiencies persist it is driven by behavioural biases, liquidity constraints, and structural breaks—the long-term trajectory of Indian markets points toward increased resilience and rational pricing.

Future research could extend these findings by employing high-frequency intraday data, sector-specific models, or international comparative analysis. Moreover, integrating macroeconomic variables and market sentiment indicators would further enrich understanding of anomaly dynamics in complex market environments.

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8. Appendix:

Fig 1.A : F-test output BSE

(1) Monday = 0
(2) Tuesday = 0
(3) Wednesday = 0
(4) Thursday = 0
F(4, 1230) = 3.39
Prob > F = 0.0091

Fig 2.A : F-test output NSE

(1) Monday = 0
(2) Tuesday = 0
(3) Wednesday = 0
(4) Thursday = 0
F(4, 1231) = 3.49
Prob > F = 0.0077

Fig 3.A : BSE Breusch Godfrey Test outputs

Breusch Godfrey LM test for autocorrelation chi2	df	Prob>Chi2
23.5640	5	0.0003

Fig 4.A : NSE Breusch Godfrey Test outputs

Breusch Godfrey LM test for autocorrelation chi2	df	Prob>Chi2
25.3350	5	0.0001

Fig 5.A : NSE stationary series plot of returns over full sample

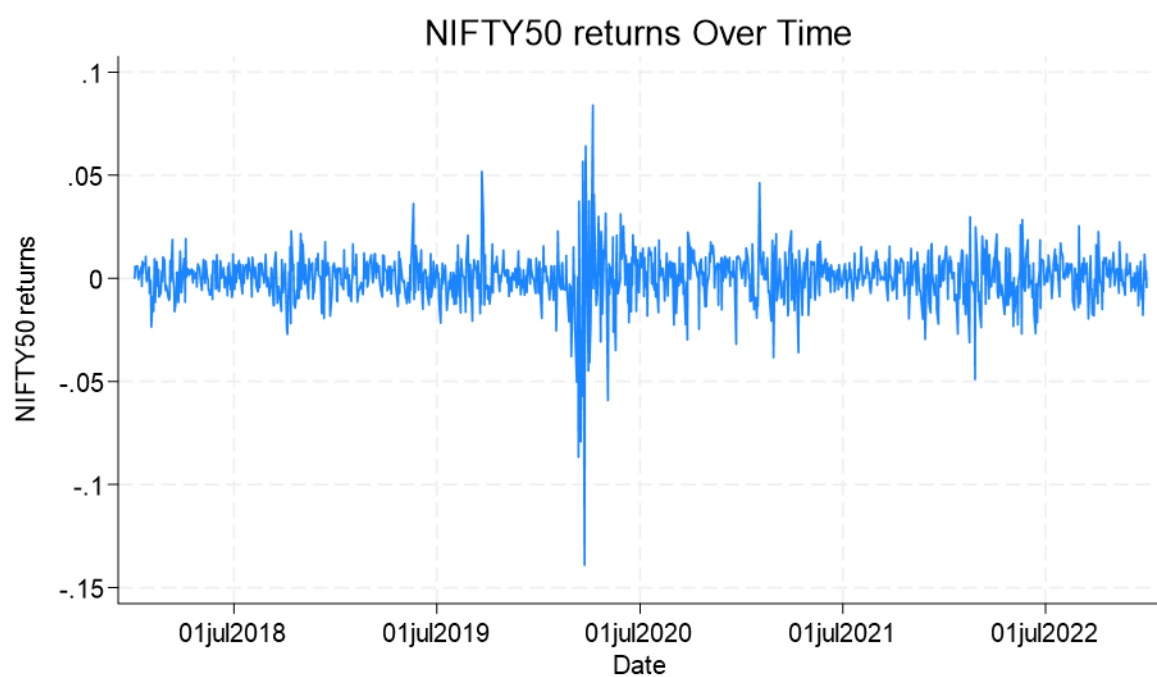


Fig 6.A : NSE stationary series plot of returns over pre-covid sample

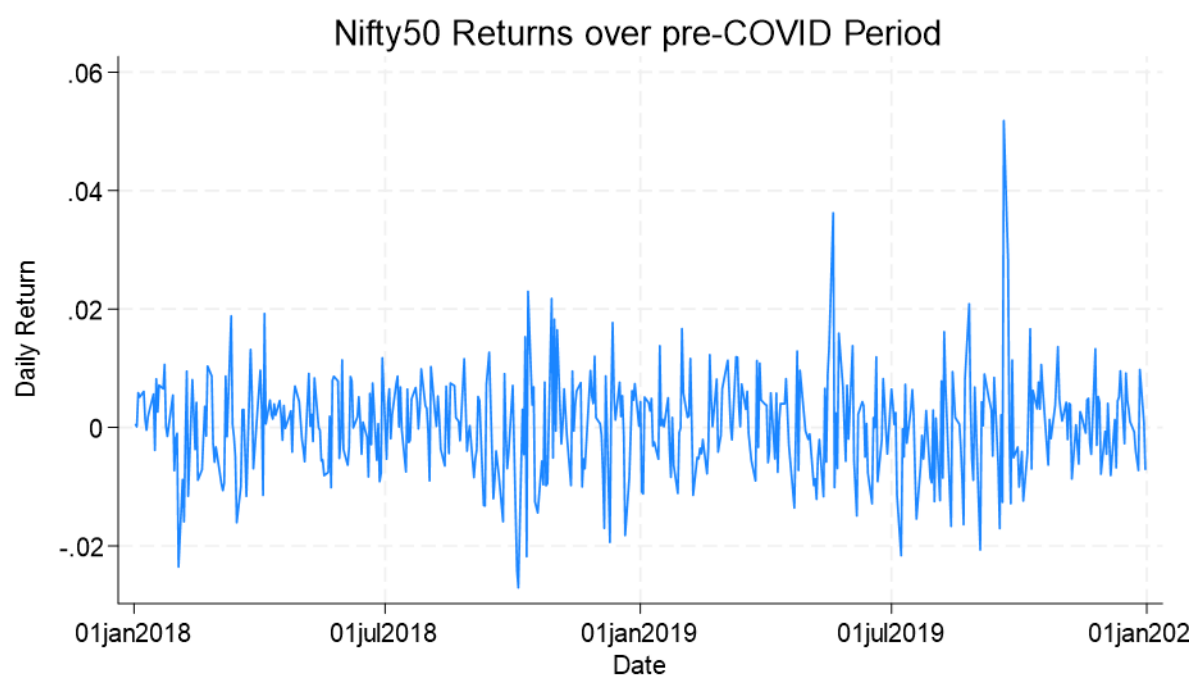


Fig 7.A : NSE stationary series plot of returns during covid sample

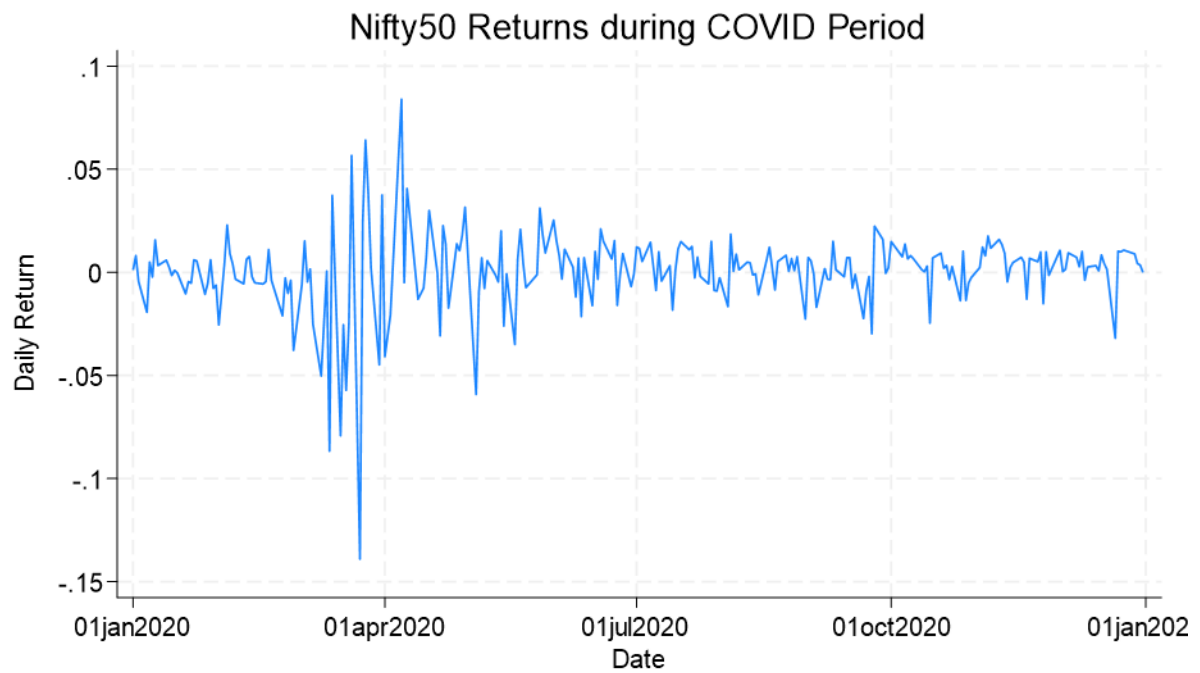


Fig 8.A : NSE stationary series plot of returns post-covid sample

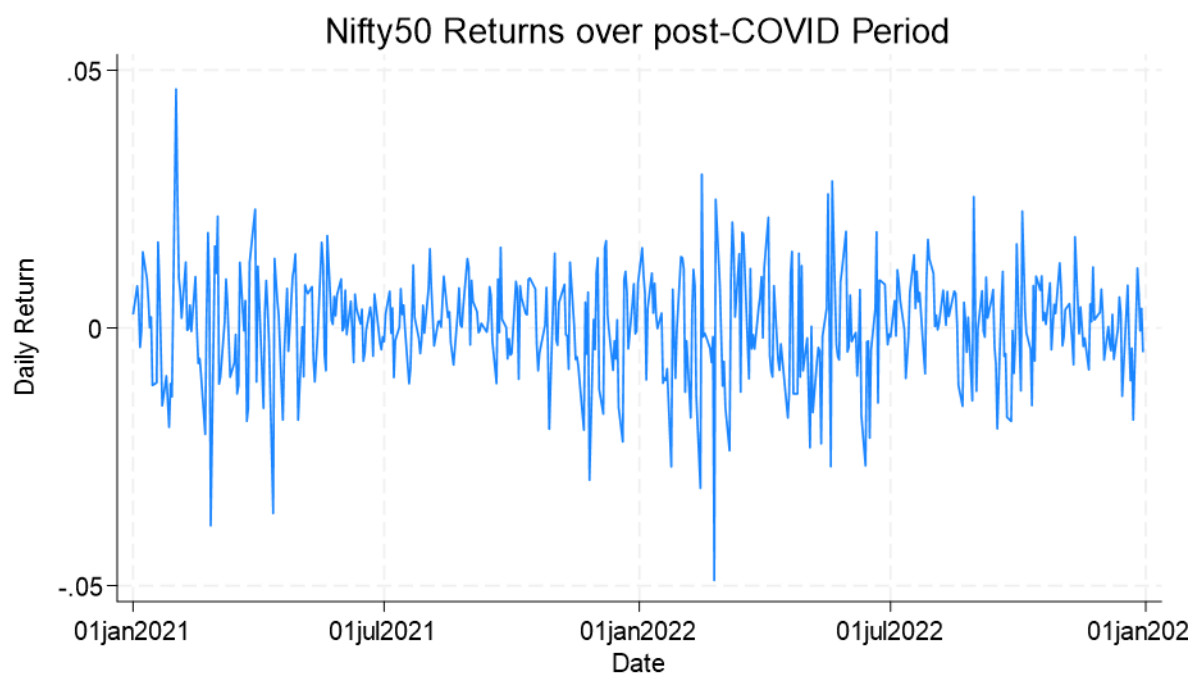


Fig 9.A : BSE stationary series plot of returns over full sample

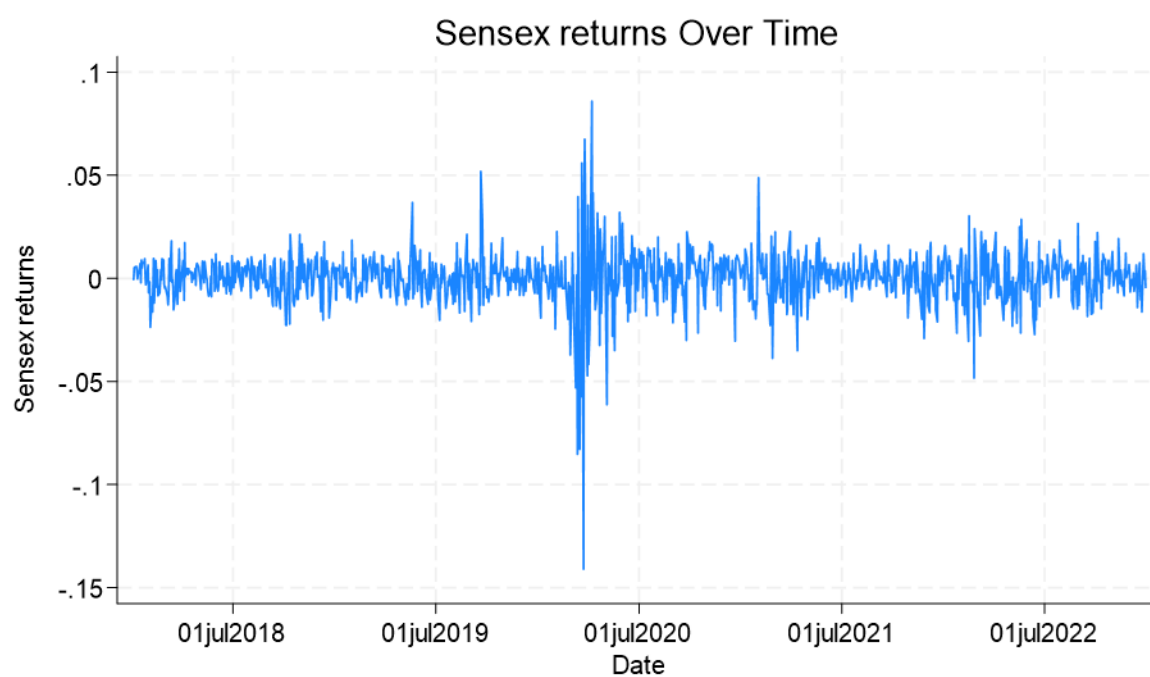


Fig 10.A : BSE stationary series plot of returns over pre-covid sample

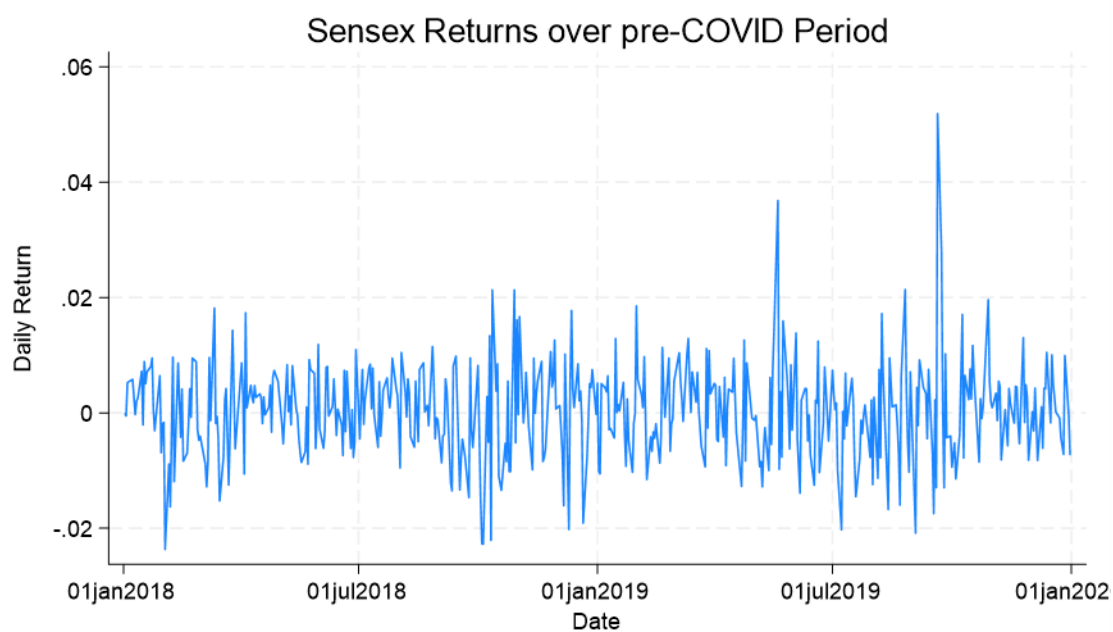


Fig 11.A : BSE stationary series plot of returns for during covid sample

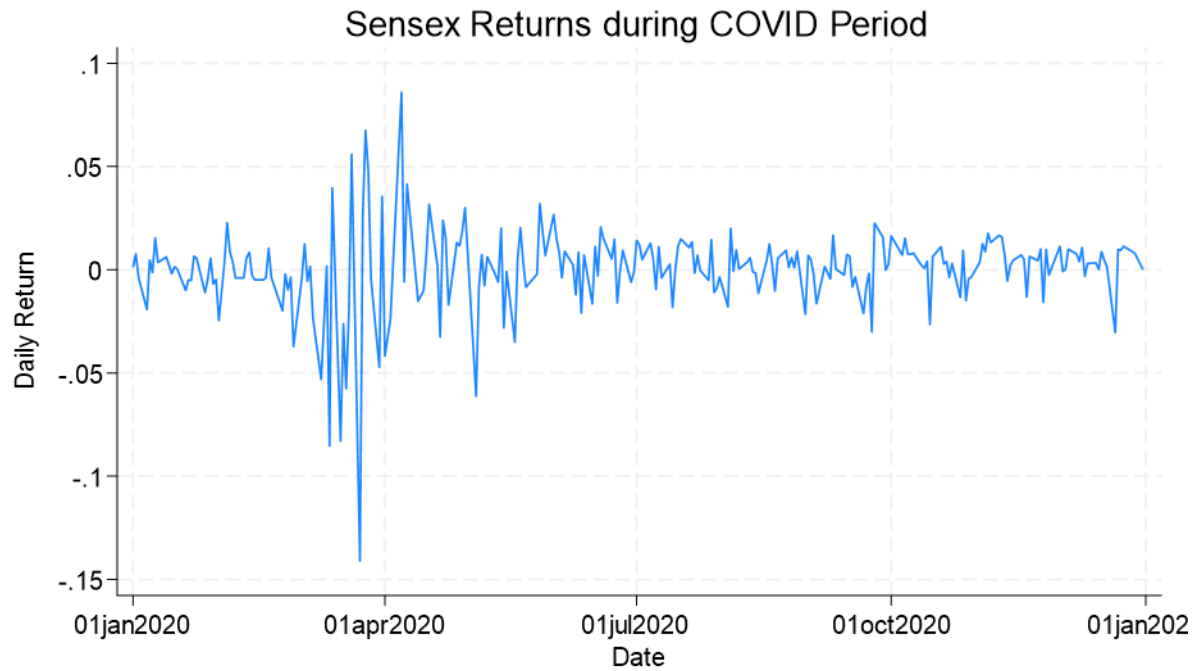


Fig 12.A : BSE stationary series plot of returns for post-covid sample

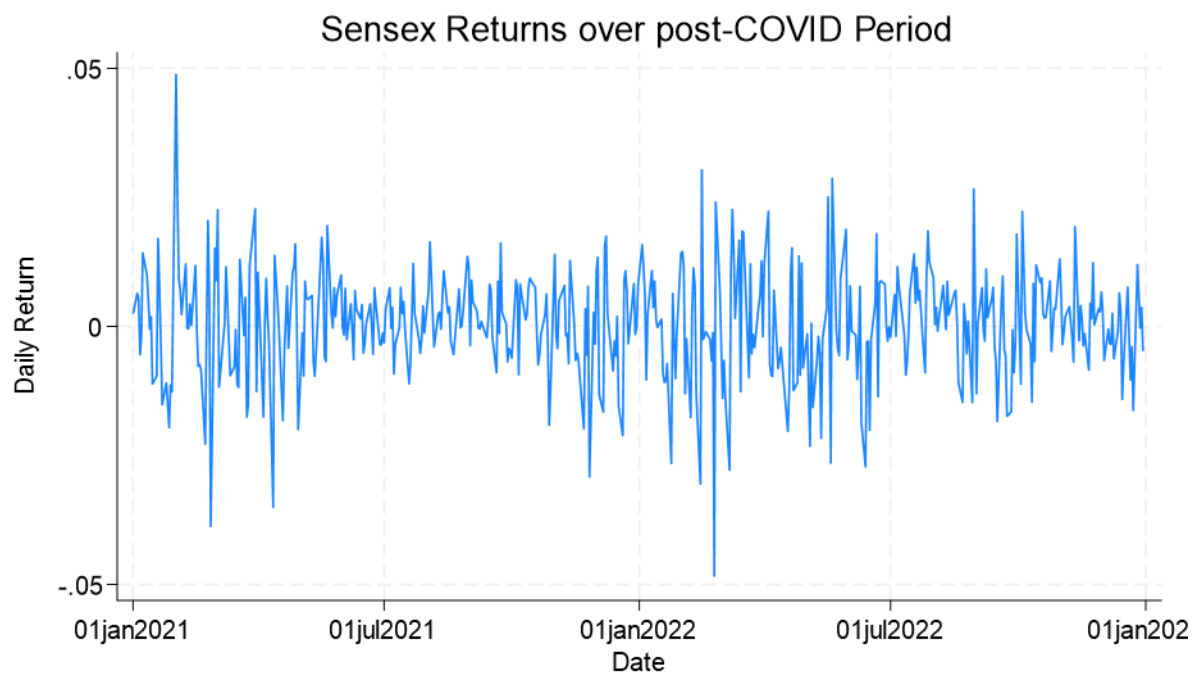


Fig 13.A : BSE's Sensex index lagged return Newey West corrected OLS regressions over subperiods (pre-covid, covid & post covid)

Regression with Newey West standard errors

return	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
L1 return	-.0443	.0428	-1.04	.3004	-.1282	.0396	
Monday	-.0018	.0013	-1.33	.1853	-.0044	.0008	
Tuesday	.0023	.0011	2.08	.0375	.0001	.0045	**
Wednesday	.0005	.001	0.45	.6493	-.0015	.0024	
Thursday	-.0002	.0011	-0.16	.8725	-.0023	.002	
pre_covid	-.0001	.0007	-0.17	.8657	-.0014	.0012	
Constant	.0004	.0008	0.50	.6177	-.0011	.0019	
Mean dependent var		0.0005	SD dependent var		0.0125		
Number of obs		1236	F-test		2.3714		

*** $p < .01$, ** $p < .05$, * $p < .1$

Regression with Newey West standard errors

return	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
L1 return	-.0443	.0427	-1.04	.3001	-.1281	.0395	
Monday	-.0018	.0013	-1.32	.1873	-.0044	.0009	
Tuesday	.0023	.0011	2.08	.0373	.0001	.0045	**
Wednesday	.0005	.001	0.46	.6487	-.0015	.0024	
Thursday	-.0002	.0011	-0.16	.8725	-.0023	.002	
covid	.0001	.0014	0.08	.9357	-.0026	.0028	
Constant	.0003	.0008	0.38	.7003	-.0013	.0019	
Mean dependent var		0.0005	SD dependent var		0.0125		
Number of obs		1236	F-test		2.3486		

*** $p < .01$, ** $p < .05$, * $p < .1$

Regression with Newey West standard errors

return	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
L1 return	-.0443	.0428	-1.03	.3013	-.1283	.0397	
Monday	-.0018	.0013	-1.32	.1865	-.0044	.0009	
Tuesday	.0023	.0011	2.08	.0377	.0001	.0045	**
Wednesday	.0005	.001	0.46	.6486	-.0015	.0024	
Thursday	-.0002	.0011	-0.16	.8738	-.0023	.002	
post_covid	0	.0007	0.06	.9561	-.0014	.0014	
Constant	.0003	.0008	0.42	.6765	-.0012	.0018	
Mean dependent var		0.0005	SD dependent var		0.0125		
Number of obs		1236	F-test		2.2227		

*** $p < .01$, ** $p < .05$, * $p < .1$

Fig 14.A : NSE's Nifty50 index lagged return Newey West corrected OLS regressions over subperiods (pre-covid, covid & post covid)

Regression with Newey West standard errors

return	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
L1 return	-.0438	.044	-1.00	.3198	-.13	.0425	
Monday	.0036	.0006	6.30	0	.0025	.0048	***
Tuesday	-.0017	.0012	-1.46	.1448	-.004	.0006	
Wednesday	.0024	.0009	2.74	.0062	.0007	.0041	***
Thursday	.0005	.0008	0.66	.5071	-.001	.002	
pre_covid	-.0002	.0007	-0.34	.7316	-.0016	.0011	
Constant	.0003	.0006	0.52	.6039	-.0009	.0015	
Mean dependent var		0.0004	SD dependent var			0.0123	
Number of obs		1237	F-test			18.6977	

*** $p < .01$, ** $p < .05$, * $p < .1$

Regression with Newey West standard errors

return	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
L1 return	-.0437	.044	-0.99	.3207	-.13	.0426	
Monday	.0035	.0005	6.89	0	.0025	.0045	***
Tuesday	-.0017	.0012	-1.45	.1465	-.004	.0006	
Wednesday	.0024	.0009	2.74	.0062	.0007	.0041	***
Thursday	.0005	.0008	0.66	.5078	-.001	.002	
covid	.0001	.0014	0.08	.9333	-.0026	.0028	
Constant	.0002	.0005	0.41	.6848	-.0008	.0012	
Mean dependent var		0.0004	SD dependent var			0.0123	
Number of obs		1237	F-test			20.8043	

*** $p < .01$, ** $p < .05$, * $p < .1$

Regression with Newey West standard errors

return	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
L1 return	-.0437	.0442	-0.99	.3221	-.1304	.0429	
Monday	.0036	.0006	5.81	0	.0024	.0048	***
Tuesday	-.0017	.0012	-1.46	.1447	-.004	.0006	
Wednesday	.0024	.0009	2.74	.0062	.0007	.0041	***
Thursday	.0005	.0008	0.67	.5057	-.001	.002	
post_covid	.0002	.0007	0.22	.8283	-.0012	.0015	
Constant	.0002	.0006	0.28	.7813	-.001	.0014	
Mean dependent var		0.0004	SD dependent var			0.0123	
Number of obs		1237	F-test			15.4244	

*** $p < .01$, ** $p < .05$, * $p < .1$